An uncomputable phase diagram at positive temperature for one classical lattice model

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Abstract

Can the phase diagram of one fixed classical lattice model be uncomputable? For quantum spin systems at zero temperature it can: there are families of Hamiltonians whose spectral gap or ground-state phase cannot be computed from the parameter. We prove a classical counterpart at positive temperature. There is one translation-invariant nearest-neighbour interaction on $\mathbb{Z}^2$, with local energies 0, 1 and $+\infty$ (hard constraints) and a fixed universal Turing machine U built in, and there are disjoint intervals J_n with rational endpoints computable from n, such that at every inverse temperature β in J_n the model has exactly one translation-invariant Gibbs measure if U does not halt on input n, and at least two if U halts. So uniqueness of the translation-invariant Gibbs measure at rational inverse temperature is undecidable for this single model. Uniqueness among all Gibbs measures fails at every temperature, because the hard constraints freeze a non-periodic background.

The background is a hierarchical tiling that contains arbitrarily large square grids. On each grid a Turing machine reads a label n from the size of the grid and runs U on n. Where U halts, small sensors made of free bits and unit energies couple the corner spins of all grids of that size into an ordinary Ising model, whose coupling is large only when β is close to $n \log 2$. Assuming the exact solution of the two-dimensional Ising model, the inverse temperatures with at least two translation-invariant Gibbs measures form exactly a union of explicit intervals around $n \log 2$ over the halting inputs n; restricted to rational β, this set is Σ_1^0-complete.

AI systems assisted with this work. The authors take responsibility for its claims; see "How this work was done." This preprint has not been peer reviewed.

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1 Introduction

1.1 Can one model have an uncomputable phase diagram?

A classical lattice model assigns a finite set of local states to each site of $\mathbb{Z}^d$ and an energy to local patterns. Its equilibrium states at inverse temperature β are its Gibbs measures, and its phase diagram records the values of β at which the Gibbs measure is unique and those at which several phases coexist. For the two-dimensional Ising model the diagram is a single threshold, which Onsager computed exactly [27]. Can the phase diagram of one fixed model, with a finite description, be impossible to compute?

For quantum spin systems at zero temperature the answer is yes. Cubitt, Perez-Garcia and Wolf proved that the spectral gap of translation-invariant nearest-neighbour Hamiltonians on a two-dimensional lattice is undecidable [9, 10], and the same holds in one dimension [2]. Bausch, Cubitt and Watson then built one continuous family of such Hamiltonians $H(\varphi)$ whose phase cannot be computed from φ: on an interval of φ starting at 2^{-n}, the Hamiltonian is gapless with a critical ground state if a Turing machine halts on n, and gapped with a unique product ground state if it does not [3]. For classical models, Gu, Weedbrook, Perales and Nielsen showed that ground-state observables of two-dimensional Ising lattices built from Turing machines are undecidable, and expected the result to extend to positive temperature: it "readily generalizes to lattices of finite temperatures" [16]. Gayral, Sablik and Taati wrote: "We expect that, in two or more dimensions, most questions regarding the thermodynamic behaviour of finite-range models at or near zero temperature are undecidable" [14].

This paper gives one fixed classical model, at positive temperature, whose phase diagram encodes the halting problem. It is a classical, positive-temperature counterpart of the quantum results above, with the temperature in the role of the parameter φ. The model has hard constraints: some local patterns have energy $+\infty$ and never occur.

Theorem A (Theorem 7.3). Fix a universal Turing machine U. There are a finite set $\mathfrak{S}$ and a translation-invariant nearest-neighbour interaction Φ on $\mathfrak{S}^{\mathbb{Z}^2}$, with every local energy in $\{0, 1, +\infty\}$ and with U hard-coded, such that the following holds. Put $\lambda_K = 2\sum_{j=0}^{K-1} \frac{1}{(2j+1)3^{2j+1}}$, $\beta_n = n\lambda_{n+3}$ and $J_n = (\beta_n - \frac{1}{1000}, \beta_n + \frac{1}{1000})$ for $n \geq 1$; these are pairwise disjoint intervals of positive numbers with rational endpoints computable from n. For every $n \geq 1$ and every real $\beta \in J_n$:

(i) if U does not halt on input n, then $\beta\Phi$ has exactly one translation-invariant Gibbs measure;

(ii) if U halts on input n, then $\beta\Phi$ has at least two translation-invariant Gibbs measures.

At every $\beta > 0$, $\beta\Phi$ also has Gibbs measures that are not translation invariant: uniqueness in (i) is uniqueness among translation-invariant Gibbs measures, and uniqueness among all Gibbs measures fails at every β.

Gibbs measures are taken with the standard convention for hard constraints: they are supported on the configurations of finite energy, and their conditional laws are Boltzmann

weights over the admissible fillings of finite volumes (Definition 2.2). The partial sums λ_K of the series $\log 2 = 2\sum_{j\geq 0} \frac{1}{(2j+1)3^{2j+1}}$ make β_n rational and place it within $3/26244$ of $n \log 2$ (Lemma 7.1).

Corollary B (Corollary 7.4). The set of rational $\beta > 0$ at which $\beta\Phi$ has exactly one translation-invariant Gibbs measure is undecidable: the rational number β_n, computable from n, belongs to it if and only if U does not halt on n.

With the exact solution of the two-dimensional Ising model as an additional input, the whole diagram can be described: the coexistence set is a union of explicit intervals around $n \log 2$, one for each n on which U halts (Proposition 8.2).

1.2 How it works

The interaction has two layers, drawn in Figure 1. The *background* is frozen by the hard constraints. The *payload* carries Ising spins and fluctuates at temperature β.

- *A frozen hierarchical tiling.* The bottom of the background is Ollinger's 2×2 substitution tiling [26], in the transcription of [21], decorated with ruler bits [12]. It has exactly one translation-invariant probability measure, and almost every tiling under it is rigid: no change of finitely many tiles gives another tiling (Proposition 3.7).
- *Grids at every depth.* The light red squares of the tiling are nested frames. Following Robinson's method [29], the frames of depth d cut out square grids of $N_d \times N_d$ sites, $N_d = 2^{d+2} - 2$, joined by straight channels (Proposition 5.1). Every depth occurs with positive density.
- *A computation on each grid.* A static binary counter on each grid finds d. A fixed Turing machine M computes the *label* $n(d) = 1 + v_2(d)$, where v_2 is the 2-adic valuation, and runs U on $n(d)$ inside the grid. The grid is *ready* if this computation halts before the grid ends (Lemma 5.5). Every label is the label of infinitely many depths, so if U halts on n, every sufficiently large depth with label n is ready.
- *Ising layers from sensors.* Every site carries a free spin. Wires join the spin at the south-west corner, the *root*, of each grid to the roots of the next grids of the same depth. At a ready root two *sensors* compare the root spin with the incoming wires. A sensor is a finite gadget of free bits and unit energies; summing over it gives exactly an Ising coupling $J_n(\beta)$ between neighbouring roots, with $e^{2J_n} = 1 + 1/b_n$ and

  $$b_n(\beta) = 2^{-4} \cosh\big(20(\beta - n \log 2)\big) + 2^{-10}$$

  (Lemma 6.2). The coupling is strong ($e^{J_n} > 3$) when $|\beta/\log 2 - n| < 1/20$ and very weak at distance at least $19/20$.

So for $\beta \in J_n$ the payload is a disjoint union of Ising models, one for each ready depth. All of them are in the high-temperature regime of Dobrushin's uniqueness theorem unless some ready depth has label n, which happens exactly when U halts on n; then the Peierls argument gives two phases. Integrating the payload over the frozen background turns these statements into statements about translation-invariant Gibbs measures of $\beta\Phi$ (Section 7).

1.3 Objections answered

The hard constraints. The model is a hard-constraint model, and the infinite energies are essential to the proof. They freeze the background, so that finite-volume conditional laws

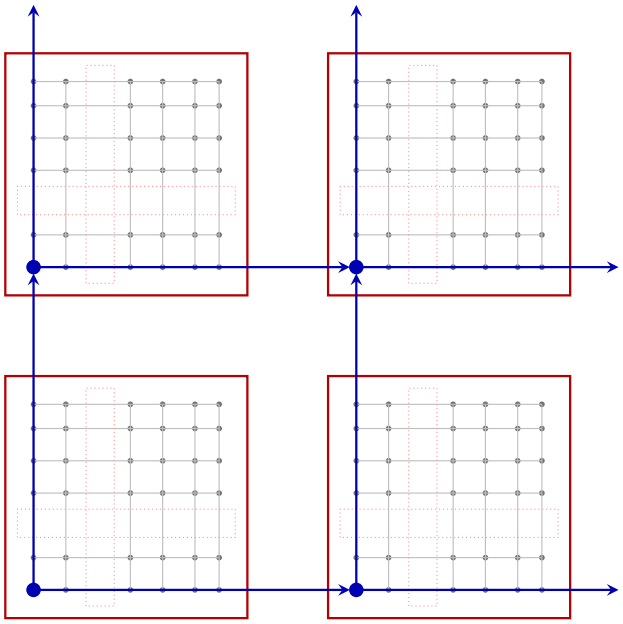


Figure 1: The two layers of the model, schematically. Inside each frame of depth d, the sites whose row and column avoid all lower frames form an $N_d \times N_d$ grid, on which the machine M runs. The roots of all grids of depth d form a square lattice of spacing 4^{d+1}. Each root sends its spin east and north along wires; when the grids of depth d are ready, sensors at the receiving roots turn these wires into the edges of an Ising model with coupling $J_{n(d)}(\beta)$.

cannot change the tiling, the grids or the computations, and they keep every computation exact. Nothing in this paper applies to interactions with finite energies only. For finite-range interactions with computable finite energies, the pressure is a computable function of β when the configuration space is a full shift [14, Proposition 14]. A finite-energy version of Theorem A is the first open problem of Section 9.

Translation invariance. Each typical frozen background, with the payload in equilibrium on it, is a Gibbs measure that is not translation invariant. So at every β there are uncountably many Gibbs measures, and the statement about uniqueness concerns the translation-invariant ones; Theorem A states this. In case (i), the unique translation-invariant Gibbs measure is the average of the payload equilibria over the unique translation-invariant law of the background.

Is the input hidden in the interaction? No. The interaction is one finite table, fixed before n. The intervals J_n are computable from n without knowing whether U halts. The halting of U on n enters only through which grids of the frozen background complete their computation.

Probabilistic cellular automata. Kurdyumov [20] and Toom [31] proved that ergodicity of probabilistic cellular automata is undecidable, for automata in which some transitions have probability zero; see also [6]. The space–time histories of such an automaton form a lattice system one dimension up with hard constraints, so these results already concern classes of hard-constraint systems, with one system for each machine. In [12] we showed that ergodicity remains undecidable for a computable family of automata with positive rates, all at one fixed noise rate, and deduced that uniqueness of the translation-invariant Gibbs measure is undecidable for the associated family of space–time interactions; there too the machine is part of the model. Here the interaction is fixed and the input is carried by the temperature.

1.4 Prior work

Table 1 places Theorem A among the closest results. Miękisz and Taati built a finite-range lattice gas on $\mathbb{Z}^4$ whose low-temperature Gibbs states are perturbations of Ammann tilings, by running a noise-resilient computation inside its Gibbs states [23]. Computability also enters thermodynamics through zero-temperature limits of Gibbs measures [7] and through the entropies of multidimensional subshifts of finite type [18]. Marsan, Sablik and Törmä built a perturbed cellular automaton whose ergodicity undergoes at least two transitions as the noise varies, and asked which sets of noise parameters can occur as ergodicity sets [22]. The scaffold uses Wang tiles [32], the hierarchical method of Berger [5] and Robinson [29], and substitution tilings with local rules and unique derivation [25, 24, 30], through Ollinger's substitution [26] as transcribed in [21]; the ruler decoration and its proofs are from [12].

Result	System	Temperature	One fixed model	Local terms
Undecidable spectral gap [9, 10]	quantum	zero	no	finite
Uncomputable phase diagram [3]	quantum	zero	yes, parameter φ	finite
Undecidable ergodicity of automata [20, 31]	classical	dynamics	no	some zero
Undecidable ground-state observables [16]	classical	zero	no	finite
Undecidable chaoticity [14]	classical	zero (limit)	no	finite
Undecidable ergodicity at fixed noise [12]	classical	positive	no	finite
Theorem A	classical	positive, parameter β	yes	$\{0, 1, +\infty\}$

Table 1: Undecidability of phases. "One fixed model" means that the interaction is fixed and the input enters only through a real parameter. "Some zero" means that some transition probabilities vanish; for [12] the local weights are the positive transition probabilities of the automata.

1.5 Reading guide

Section 2 fixes the definitions: interactions with hard constraints, Gibbs measures, Turing machines, and two classical facts about the Ising model. Section 3 presents the scaffold and proves that it has one translation-invariant law and is rigid; the finite-table proofs about the substitution are in Section A. Section 4 defines the interaction completely: every field, every local constraint and every energy term. Its tables are collected in Section B, and the machine M is given in Section C. Section 5 shows that the background is determined by the scaffold and computes it. Section 6 eliminates the sensors and identifies the payload with a disjoint union of Ising models. Section 7 proves Theorem A and Corollary B. Section 8 gives the exact diagram, conditional on the classical solution of the Ising model, and Section 9 lists open problems. A reader interested only in the thermodynamics can take Proposition 5.7 and Lemma 5.8 and Corollary 6.4 as black boxes and read Section 7.

2 Setting

Throughout, $\mathbf{e}_1 = (1,0)$ and $\mathbf{e}_2 = (0,1)$. Integer intervals are sets of integers, $[a,b] = \{a,\dots,b\}$ and $[0,m) = \{0,\dots,m-1\}$, and $z \bmod m \in [0,m)$. For $a \in \mathbb{Z}^2$ the shift T_a acts on configurations by $(T_a\omega)(\xi) = \omega(\xi - a)$ and on measures by $T_a\mu = \mu \circ T_a^{-1}$.

2.1 Interactions with hard constraints and Gibbs measures

Let $\mathfrak{S}$ be a finite set and $\Omega = \mathfrak{S}^{\mathbb{Z}^2}$, with the product topology and its Borel σ-algebra. For $\Lambda \subseteq \mathbb{Z}^2$ write ω_Λ for the restriction of ω to Λ, and $\mathcal{F}_\Lambda$ for the σ-algebra generated by the coordinates in Λ.

Definition 2.1 (Interactions)**.** An *interaction* is a family $\Phi = (\Phi_A)$, indexed by the finite sets $A \subseteq \mathbb{Z}^2$, of functions $\Phi_A \colon \Omega \to [0,+\infty]$ such that $\Phi_A(\omega)$ depends only on ω_A. It is *translation invariant* if $\Phi_{A+a}(T_a\omega) = \Phi_A(\omega)$ for all A, a and ω, and *nearest-neighbour* if $\Phi_A = 0$ unless A is a single site or a pair $\{\xi, \xi + \mathbf{e}_i\}$. A configuration ω is *legal* if $\Phi_A(\omega) < \infty$ for every A; the set of legal configurations is X_Φ. For a finite set Λ, the energy of ω in Λ is $H_\Lambda(\omega) = \sum_{A \cap \Lambda \neq \emptyset} \Phi_A(\omega) \in [0,+\infty]$.

For a nearest-neighbour interaction the sum defining H_Λ is finite. If ω is legal and η agrees with ω outside Λ, then η is legal exactly when $H_\Lambda(\eta) < \infty$, because the terms not meeting Λ are the same for η and ω. We use $e^{-\infty} = 0$.

Definition 2.2 (Gibbs measures)**.** Let Φ be a nearest-neighbour interaction and $\beta > 0$. For a finite $\Lambda \subseteq \mathbb{Z}^2$ and a legal ω, the *Gibbs kernel* is the probability measure on Ω

$$\gamma_\Lambda(\eta \mid \omega) = \frac{\mathbf{1}[\eta_{\Lambda^c} = \omega_{\Lambda^c}]\, e^{-\beta H_\Lambda(\eta)}}{Z_\Lambda(\omega)}, \qquad Z_\Lambda(\omega) = \sum_{\eta \colon \eta_{\Lambda^c} = \omega_{\Lambda^c}} e^{-\beta H_\Lambda(\eta)}.$$

It is well defined, since $Z_\Lambda(\omega) \geq e^{-\beta H_\Lambda(\omega)} > 0$, and it is supported on legal configurations. For a bounded measurable F write $(\gamma_\Lambda F)(\omega) = \sum_\eta \gamma_\Lambda(\eta \mid \omega) F(\eta)$. A *Gibbs measure for* $\beta\Phi$ is a probability measure μ on Ω with $\mu(X_\Phi) = 1$ such that

$$\int F \, d\mu = \int (\gamma_\Lambda F)\, d\mu$$

for every finite Λ and every bounded measurable F. It is *translation invariant* if $T_a\mu = \mu$ for all $a \in \mathbb{Z}^2$.

This is the standard convention for interactions with hard constraints: Gibbs measures live on the legal configurations, and the kernels sum over the admissible fillings of Λ only. Kernels are never evaluated at illegal boundary conditions. Since $\gamma_\Lambda F$ is $\mathcal{F}_{\Lambda^c}$-measurable when F is $\mathcal{F}_\Lambda$-measurable, the identity above is equivalent to $\mathbb{E}_\mu[F \mid \mathcal{F}_{\Lambda^c}] = \gamma_\Lambda F$ μ-almost surely for bounded measurable F.

We shall also use specifications whose variables are not indexed by sites. Let $\mathcal{V}$ be a countable set of *coordinates*, each with a finite set of values, and let a family of nonnegative *factors* be given, each a function of finitely many coordinates, with each coordinate in finitely many factors. For a finite $S \subseteq \mathcal{V}$ and a configuration p of all coordinates, the kernel $\gamma_S(\cdot \mid p)$ is the probability on configurations agreeing with p off S proportional to the product of the factors that involve a coordinate of S, whenever the normalization is positive. A probability measure on configurations at which every factor is positive is *Gibbs* for these factors if it satisfies the identity of Definition 2.2 for every finite $S \subseteq \mathcal{V}$.

Lemma 2.3 (Consistency)**.** *In the setting of factors above, let $S \subseteq S'$ be finite sets of coordinates, and let μ be a probability measure on configurations at which every factor is positive. If μ satisfies the Gibbs identity for S', it satisfies it for S.*

Proof. Fix p with positive factors. The kernel $\gamma_{S'}(\cdot \mid p)$ is proportional to the product of the factors meeting S'; conditioning it on the values of the coordinates in $S' \setminus S$ leaves a measure proportional to the product of the factors meeting S, because the remaining factors are then constant. So $\gamma_{S'}(\gamma_S F) = \gamma_{S'} F$ for bounded F. Hence $\int F\,d\mu = \int \gamma_{S'} F\,d\mu = \int \gamma_{S'}(\gamma_S F)\,d\mu = \int \gamma_S F\,d\mu$. □

For an interaction on sites, the coordinates are the fields at the sites (Section 4); the kernel of a set Λ of sites is the kernel of the set of all coordinates at the sites of Λ. By Lemma 2.3, a measure that is Gibbs for all finite sets of sites is Gibbs for all finite sets of coordinates, since every finite set of coordinates lies in the set of coordinates of a finite set of sites; the converse is immediate.

2.2 Turing machines

A deterministic Turing machine has finitely many states, among them an initial state and halting states, a finite tape alphabet with a blank symbol, and a transition function. Its tape has cells $0, 1, 2, \ldots$, its head starts on cell 0, and at each step it writes a symbol, changes state and moves one cell left or right. We use the standard convention that the machine never moves left from cell 0; every machine can be converted into one with this property and the same halting behaviour. A machine *halts on input* $n \geq 1$ if, started on the tape whose cells $0, \ldots, n-1$ hold the symbol 1 and whose other cells are blank, it enters a halting state.

We fix once and for all a deterministic Turing machine U of this kind that is universal: its halting set $\mathcal{H}_U = \{n \geq 1 : U \text{ halts on } n\}$ is Σ_1^0-complete, in particular undecidable. Its transition table is part of the interaction of Section 4. If U halts on n, write $t_U(n)$ for the number of steps it makes and $s_U(n)$ for the largest cell its head visits.

2.3 Two facts about the Ising model

Let $G = (V, E)$ be a countable graph in which every vertex has at most four neighbours, and let $J \colon E \to [0, \infty)$. The *Ising model* on G with couplings J has spins $\sigma \in \{-1, +1\}^V$ and the factors $\exp(J_{uv}\sigma_u\sigma_v)$, one for each edge uv. On $\mathbb{Z}^2$ with all couplings equal to J we speak of the Ising model with coupling J.

Lemma 2.4 (Peierls)**.** *For every $J \geq 0$, the Ising model on $\mathbb{Z}^2$ with coupling J has Gibbs measures ν^+ and ν^-, the plus and minus states (the weak limits of the finite-volume measures with plus and minus boundary conditions), invariant under all translations of $\mathbb{Z}^2$ and exchanged by the global spin flip. If $e^J \geq 3$, then $\nu^+(\sigma_0 = -1) \leq 1/3$ and $\nu^-(\sigma_0 = +1) \leq 1/3$; in particular $\nu^+ \neq \nu^-$.*

Proof. The plus state ν^+ is the weak limit of the finite-volume measures with plus boundary condition on boxes increasing to $\mathbb{Z}^2$; by the FKG inequality this limit exists, does not depend on the sequence of boxes, and is invariant under all translations, see [13, Chapter 3]. The minus state ν^- is its image under the global spin flip. By Peierls' argument [28], in a box with plus boundary condition the event $\sigma_0 = -1$ forces a closed contour of the dual lattice around the origin, made of edges between disagreeing spins; a given contour of length k has probability at most e^{-2Jk}, because flipping the spins inside it is injective and raises the weight by e^{2Jk}. A contour of length k around the origin crosses the positive horizontal axis within

distance $k/2$ of the origin, and starting from that crossing it is a closed walk of k steps with at most three choices at each step after the first, so there are at most $k\,3^k$ of them; $k \geq 4$. Hence

$$\nu^+(\sigma_0 = -1) \leq \sum_{k \geq 4} k\,3^k e^{-2Jk} \leq \sum_{k \geq 4} k\,3^{-k} = \frac{3}{4} - \frac{2}{3} = \frac{1}{12} \leq \frac{1}{3},$$

using $3e^{-2J} \leq 1/3$ and $\sum_{k \geq 1} k3^{-k} = 3/4$. The bound passes to the limit, and the minus state is symmetric. □

Lemma 2.5 (Dobrushin)**.** *Let G be a countable graph in which every vertex has at most four neighbours, and let $J\colon E \to [0, \bar{J}]$ with $4 \tanh \bar{J} < 1$. Then the Ising model on G with couplings J has exactly one Gibbs measure.*

Proof. The conditional law of σ_u given the other spins is $\mathbb{P}(\sigma_u = +1) = \frac{1}{2}(1 + \tanh h)$ with $h = \sum_{v \sim u} J_{uv}\sigma_v$. Changing one neighbour σ_v changes h by $2J_{uv}$, and so changes this probability by at most $\sup_t \frac{1}{2}\big(\tanh(t + J_{uv}) - \tanh(t - J_{uv})\big) = \tanh J_{uv}$. Dobrushin's interdependence matrix therefore has row sums at most $4 \tanh \bar{J} < 1$, and Dobrushin's uniqueness theorem [11], [15, Theorem 8.7] gives exactly one Gibbs measure. Existence holds because the spin space is compact and the specification has finite range. □

A Gibbs measure in Lemma 2.5 is unique among all Gibbs measures, translation invariant or not, and the graph may be a disjoint union of infinitely many square lattices with different couplings. Both features are used in Section 7.

3 The scaffold

This section presents the bottom layer of the background: a Wang set $\mathbb{A}$ of 276 letters with a substitution of expansion 2, its quartered version $\mathbb{A}^{\mathrm{q}}$, and the light red squares that will serve as frames. The base set and the ruler decoration are those of the scaffold of [12]; we restate the facts we need, with proofs of the finite-table statements in Section A. Beyond those facts we prove that every translation-invariant law has Haar-distributed phase, that $\mathbb{A}$ and $\mathbb{A}^{\mathrm{q}}$ have exactly one translation-invariant probability measure, and that typical tilings are rigid (Propositions 3.7 and 3.10).

3.1 Wang sets

A *Wang set* is a finite set of tiles, each with a north, east, south and west colour, written with subscripts $\mathrm{N}, \mathrm{E}, \mathrm{S}, \mathrm{W}$. A *pattern* on $V \subseteq \mathbb{Z}^2$ is a map $P\colon V \to R$; it is *legal* if $P(v)_{\mathrm{E}} = P(v + \mathbf{e}_1)_{\mathrm{W}}$ and $P(v)_{\mathrm{N}} = P(v + \mathbf{e}_2)_{\mathrm{S}}$ whenever both sites lie in V. A *tiling* is a legal pattern on $\mathbb{Z}^2$, and X_R is the set of tilings of R, a closed shift-invariant subset of $R^{\mathbb{Z}^2}$.

3.2 Ollinger's substitution and the ruler decoration

A tile of the base set is a triple (θ, κ, ψ) of a *thin* type $\theta \in \{\mathsf{a}, \mathsf{b}, \mathsf{c}, \mathsf{d}\}$, a *thick* type κ and a *black* type $\psi \in \{\mathsf{A}, \mathsf{B}, \mathsf{C}, \mathsf{D}, \mathsf{E}\}$. A thick type is one of four corners $\angle\mathsf{a}, \angle\mathsf{b}, \angle\mathsf{c}, \angle\mathsf{d}$ or one of sixteen lines $\langle \alpha, \alpha' \rangle$ with a vertical atom $\alpha \in \{\mathsf{r}_0, \mathsf{r}_2, \mathsf{g}_0, \mathsf{g}_2\}$ and a horizontal atom $\alpha' \in \{\mathsf{r}_1, \mathsf{r}_3, \mathsf{g}_1, \mathsf{g}_3\}$; the letter r means red and g green. Each component has an endpoint ∂_X on each side $X \in \{\mathrm{N}, \mathrm{E}, \mathrm{S}, \mathrm{W}\}$, listed in Table 2.

Definition 3.1 (The base set and its substitution)**.** The base set $\mathbb{O}$ is the Wang set of the 104 triples of Table 3; the colour of (θ, κ, ψ) on side X is $(\partial_X\theta, \partial_X\kappa, \partial_X\psi)$. For a triple $t = (\theta, \kappa, \psi)$, the pattern $\varphi(t)$ on $[0,2)^2$ is given by Table 4 (left), with the maps F, se and nw of Section A. For a pattern w on V, $\varphi(w)$ is the pattern on $2V + [0,2)^2$ with $\varphi(w)(2v + \varsigma) = \varphi(w(v))(\varsigma)$.

A *red crossing* is a line whose two atoms are red.

Definition 3.2 (The scaffold $\mathbb{A}$)**.** $\mathbb{A}$ is the set of tuples $\mathfrak{a} = (\theta, \kappa, \psi, \chi_1, \chi_2)$ with $(\theta, \kappa, \psi) \in \mathbb{O}$ and $\chi_1, \chi_2 \in \{0,1\}$ such that: (R1) if $\theta \in \{\mathsf{b}, \mathsf{c}\}$ then $\chi_1 = 0$; (R2) if $\theta \in \{\mathsf{b}, \mathsf{d}\}$ then $\chi_2 = 0$; (R3) if κ is a corner then $\chi_1 = \chi_2$; (R4) if κ is a red crossing then $\chi_1 \neq \chi_2$. The north and south colours of $\mathfrak{a}$ are those of (θ, κ, ψ) together with the *column bit* χ_1; the east and west colours are those of (θ, κ, ψ) together with the *row bit* χ_2. The decorated substitution $\varphi'(\mathfrak{a})$ is the pattern on $[0,2)^2$ whose bases are $\varphi(\theta, \kappa, \psi)$ and whose bits are given by Table 4 (right); it acts on patterns blockwise.

Table 5 counts $|\mathbb{A}| = 276$. Letters matching vertically have the same column bit, and letters matching horizontally have the same row bit, so in a tiling χ_1 is constant on columns and χ_2 on rows.

Proposition 3.3 (Desubstitution)**.** *The set $\mathbb{A}$ tiles. The map φ' is injective, and for every pattern w of $\mathbb{A}$, $\varphi'(w)$ is legal if and only if w is legal. For every tiling f of $\mathbb{A}$ there are a unique $v \in \{0,1\}^2$ and a unique tiling g of $\mathbb{A}$ with $f(v + 2v + \varsigma) = \varphi'(g(v))(\varsigma)$ for all $v \in \mathbb{Z}^2$ and $\varsigma \in [0,2)^2$.*

This is proved in [12]; the proof is recalled in Section A. It holds for every tiling, including tilings with faults along lines.

3.3 Phases, the coincidence and the translation-invariant law

Applying Proposition 3.3 repeatedly, every tiling f of $\mathbb{A}$ has a unique *tower*: tilings $f_0 = f, f_1, f_2, \ldots$ and digits $v_0, v_1, \cdots \in \{0,1\}^2$ with

$$f_m(v_m + 2v + \varsigma) = \varphi'\big(f_{m+1}(v)\big)(\varsigma) \qquad (v \in \mathbb{Z}^2,\ \varsigma \in [0,2)^2).$$

Its *phase* is $\varpi(f) = \sum_{m \geq 0} 2^m v_m \in \mathbb{Z}_2^2$, where $\mathbb{Z}_2$ is the ring of 2-adic integers. For a site $x \in \mathbb{Z}^2$ put $x^{(0)} = x$ and let $\varsigma_m(x) \in \{0,1\}^2$ and $x^{(m+1)} \in \mathbb{Z}^2$ be defined by $x^{(m)} = v_m + 2x^{(m+1)} + \varsigma_m(x)$. Then $f_m(x^{(m)}) = \varphi'(f_{m+1}(x^{(m+1)}))(\varsigma_m(x))$, and by induction

$$x - \varpi(f) \equiv \sum_{m<l} 2^m \varsigma_m(x) \pmod{2^l} \qquad (l \geq 1), \tag{1}$$

so the $\varsigma_m(x)$ are the binary digits of $x - \varpi(f)$, coordinatewise.

Lemma 3.4 (Phases)**.** *(a) $\varpi(T_a f) = \varpi(f) + a$ for every tiling f and $a \in \mathbb{Z}^2$.*

(b) For every $l \geq 1$ there is ϱ_l such that $\varpi(f) \bmod 2^l$ is determined by the restriction of f to $[-\varrho_l, \varrho_l]^2$.

(c) For every $\varpi \in \mathbb{Z}_2^2$ there is a tiling of $\mathbb{A}$ with phase ϖ.

Proof. (a) Let (f_m, v_m) be the tower of f and write $v_0 + a = v' + 2b$ with $v' \in \{0,1\}^2$. Then $T_a f(v' + 2(v + b) + \varsigma) = f(v_0 + 2v + \varsigma) = \varphi'(T_b f_1(v + b))(\varsigma)$, so by uniqueness the tower of $T_a f$ starts with the digit v' and continues with the tower of $T_b f_1$. By induction on l, $\varpi(T_a f) \equiv v' + 2(\varpi(f_1) + b) = \varpi(f) + a \pmod{2^l}$ for every l.

(b) For $p \in [0, 2^l)^2$ with binary digits $v_0, \dots, v_{l-1}$, let X_p be the set of tilings whose phase is congruent to p modulo 2^l. By uniqueness of towers, X_p is the image of $X_{\mathbb{A}}$ under the map $g \mapsto \mathrm{sub}_{v_0} \circ \cdots \circ \mathrm{sub}_{v_{l-1}}(g)$, where $\mathrm{sub}_v(g)(v + 2v + \varsigma) = \varphi'(g(v))(\varsigma)$. This map is continuous and $X_{\mathbb{A}}$ is compact, so each X_p is compact. The 4^l sets X_p are disjoint and cover $X_{\mathbb{A}}$, so each is also open in $X_{\mathbb{A}}$. A continuous map from the compact space $X_{\mathbb{A}}$ to a finite set depends only on the restriction to some finite box.

(c) With the digits v_m of ϖ, the sets of sequences $(f_m)_{m \geq 0}$ of tilings with $f_m = \mathrm{sub}_{v_m}(f_{m+1})$ for $m < l$ are nonempty (Proposition 3.3), closed and decreasing in l; a sequence in their intersection is a tower with phase ϖ, and f_0 is the required tiling. □

Put $\mathfrak{u} = (\mathsf{b}, \angle\mathsf{d}, \mathsf{A}, 0, 0)$.

Lemma 3.5 (Coincidence)**.** *Let f be a tiling of $\mathbb{A}$ and $x \in \mathbb{Z}^2$. If $\varsigma_m(x) = \varsigma_{m+1}(x) = (1,1)$ for some m, then $f_m(x^{(m)}) = \mathfrak{u}$, and $f(x)$ is determined by $\varsigma_0(x), \dots, \varsigma_{m-1}(x)$.*

Proof. By Table 4, child $(1,1)$ of a letter $(\theta, \kappa, \psi, \chi_1, \chi_2)$ is $(\mathsf{b}, F(\theta), \mathsf{A}, 0, 0)$. So $f_{m+1}(x^{(m+1)})$ has thin type b, and $f_m(x^{(m)})$ is child $(1,1)$ of a letter of thin type b, which is $(\mathsf{b}, F(\mathsf{b}), \mathsf{A}, 0, 0) = \mathfrak{u}$. Then $f(x)$ is obtained from $\mathfrak{u}$ by taking the children $\varsigma_{m-1}(x), \dots, \varsigma_0(x)$ in turn. □

Definition 3.6 (Resolving phases)**.** A phase $\varpi \in \mathbb{Z}_2^2$ is *resolving* if for every $x \in \mathbb{Z}^2$ the binary expansion of $x - \varpi$ has two consecutive digits, at positions m and $m+1$, both equal to $(1,1)$.

For a fixed x and Haar-distributed ϖ, $x - \varpi$ is Haar distributed, so its digit pairs at positions $(2k, 2k+1)$, $k \geq 0$, are independent and uniform on 16 values; none of the first K of them is $((1,1),(1,1))$ with probability $(15/16)^K$. Hence Haar-almost every phase is resolving.

Proposition 3.7 (The translation-invariant law of $\mathbb{A}$)**.** *(a) For every resolving ϖ there is exactly one tiling f_ϖ of $\mathbb{A}$ with phase ϖ.*

(b) $X_{\mathbb{A}}$ carries exactly one translation-invariant probability measure: the image of Haar measure on $\mathbb{Z}_2^2$ under $\varpi \mapsto f_\varpi$.

(c) If f is a tiling with resolving phase and f' is a tiling that agrees with f outside a finite set, then $f' = f$.

Proof. (a) Existence is Lemma 3.4(c), and uniqueness is Lemma 3.5, by (1).

(b) Let μ be a translation-invariant probability measure on $X_{\mathbb{A}}$ and $l \geq 1$. By Lemma 3.4(a) the law of ϖ mod 2^l under μ is invariant under all translations of $(\mathbb{Z}/2^l\mathbb{Z})^2$, hence uniform. So the law of ϖ is Haar measure, which gives full measure to the resolving phases, and $f = f_{\varpi(f)}$ μ-almost surely by (a). The map $\varpi \mapsto f_\varpi$ is measurable, since $f_\varpi(x)$ is computed from finitely many digits of $x - \varpi$. A translation-invariant probability measure exists, since $X_{\mathbb{A}}$ is compact and nonempty: average the translates of a point mass over growing boxes and take a limit point.

(c) Let $l \geq 1$. Choose a such that $a + [-\varrho_l, \varrho_l]^2$ avoids the finite set. Then $T_{-a}f$ and $T_{-a}f'$ agree on $[-\varrho_l, \varrho_l]^2$, so by Lemma 3.4(b),(a) $\varpi(f) - a \equiv \varpi(f') - a \pmod{2^l}$. This holds for every l, so $\varpi(f') = \varpi(f)$, and (a) gives $f' = f$. □

3.4 Quarters and light walls

The physical lattice of the model carries quarters of letters. Let $\circ$ and $\bullet$ be two new symbols.

Definition 3.8 (Quarters)**.** The quarter set $\mathbb{A}^{\mathrm{q}}$ is the Wang set of the $4 \cdot 276$ pairs $(\varsigma, \mathfrak{a}) \in \{0,1\}^2 \times \mathbb{A}$; ς is the *quadrant* and $\mathfrak{a}$ the *parent*. The colours of $(\varsigma, \mathfrak{a})$ are

$$\begin{array}{lll} \text{east:} & (\circ, \mathfrak{a}, \varsigma_2) \text{ if } \varsigma_1 = 0, & (\bullet, \mathfrak{a}_{\mathrm{E}}, \varsigma_2) \text{ if } \varsigma_1 = 1; \\ \text{west:} & (\circ, \mathfrak{a}, \varsigma_2) \text{ if } \varsigma_1 = 1, & (\bullet, \mathfrak{a}_{\mathrm{W}}, \varsigma_2) \text{ if } \varsigma_1 = 0; \\ \text{north:} & (\circ, \mathfrak{a}, \varsigma_1) \text{ if } \varsigma_2 = 0, & (\bullet, \mathfrak{a}_{\mathrm{N}}, \varsigma_1) \text{ if } \varsigma_2 = 1; \\ \text{south:} & (\circ, \mathfrak{a}, \varsigma_1) \text{ if } \varsigma_2 = 1, & (\bullet, \mathfrak{a}_{\mathrm{S}}, \varsigma_1) \text{ if } \varsigma_2 = 0, \end{array}$$

where $\mathfrak{a}_{\mathrm{E}}, \dots$ are the colours of $\mathfrak{a}$ in $\mathbb{A}$. The *quartering* of a tiling f of $\mathbb{A}$ is $f^{\mathrm{q}}(2x + \varsigma) = (\varsigma, f(x))$.

Lemma 3.9 (Quarter encoding)**.** *For every tiling Y of $\mathbb{A}^{\mathrm{q}}$ there are a unique $\varepsilon \in \{0,1\}^2$ and a unique tiling f of $\mathbb{A}$ with $Y(\xi) = f^{\mathrm{q}}(\xi - \varepsilon)$ for all ξ; conversely every such map is a tiling of $\mathbb{A}^{\mathrm{q}}$. The* quarter phase $\tau(Y) = \varepsilon + 2\varpi(f) \in \mathbb{Z}_2^2$ *satisfies $\tau(T_a Y) = \tau(Y) + a$, and $\tau(Y) \bmod 2^l$ is determined by the restriction of Y to a finite box depending only on l.*

Proof. By the colour table, $(\varsigma, \mathfrak{a})$ and $(\varsigma', \mathfrak{a}')$ match horizontally if and only if $\varsigma' \equiv \varsigma + \mathbf{e}_1 \pmod 2$ and either $\varsigma_1 = 0$ and $\mathfrak{a}' = \mathfrak{a}$, or $\varsigma_1 = 1$ and $\mathfrak{a}, \mathfrak{a}'$ match horizontally in $\mathbb{A}$; the vertical case is the same. So in a tiling Y the quadrant of $Y(\xi)$ is congruent to $\xi - \varepsilon$ modulo 2 for one ε, the four quarters of each block $\varepsilon + 2x + [0,2)^2$ have one parent $f(x)$, and adjacent blocks have matching parents. The converse is read off the same table. For the phase, write $a + \varepsilon = \varepsilon' + 2b$ with $\varepsilon' \in \{0,1\}^2$; then $T_a Y(\xi) = (T_b f)^{\mathrm{q}}(\xi - \varepsilon')$, so $\tau(T_a Y) = \varepsilon' + 2(\varpi(f) + b) = \tau(Y) + a$ by Lemma 3.4(a). The last claim follows from Lemma 3.4(b), since ε is read off one quarter. □

A phase $\tau \in \mathbb{Z}_2^2$ of $\mathbb{A}^{\mathrm{q}}$ is *resolving* if $(\tau - \varepsilon)/2$ is resolving, where $\varepsilon \equiv \tau \pmod 2$.

Proposition 3.10 (The translation-invariant law of $\mathbb{A}^{\mathrm{q}}$)**.** *(a) For every resolving τ there is exactly one tiling Y_τ of $\mathbb{A}^{\mathrm{q}}$ with quarter phase τ, and $X_{\mathbb{A}^{\mathrm{q}}}$ carries exactly one translation-invariant probability measure, the image of Haar measure under $\tau \mapsto Y_\tau$.*

(b) If Y is a tiling of $\mathbb{A}^{\mathrm{q}}$ with resolving phase and Y' is a tiling of $\mathbb{A}^{\mathrm{q}}$ that agrees with Y outside a finite set D, then $Y' = Y$.

Proof. (a) The tiling $Y_\tau(\xi) = f_\varpi^{\mathrm{q}}(\xi - \varepsilon)$, with $\varepsilon \equiv \tau \pmod 2$ and $\varpi = (\tau - \varepsilon)/2$, has phase τ, and by Lemma 3.9 and Proposition 3.7(a) it is the only one. The rest is the proof of Proposition 3.7(b), with Lemma 3.9 in place of Lemma 3.4.

(b) Write $Y(\xi) = f^{\mathrm{q}}(\xi - \varepsilon)$ and $Y'(\xi) = f'^{\mathrm{q}}(\xi - \varepsilon')$. The quadrant of $Y'(\xi)$ is congruent to $\xi - \varepsilon'$ modulo 2, and at any site $\xi \notin D$ it equals the quadrant of $Y(\xi)$; so $\varepsilon' = \varepsilon$, and the global checker phase is not changed by a finite edit. Grouping the quarters into blocks $\varepsilon + 2x + [0,2)^2$, the parents $f'(x)$ and $f(x)$ agree unless the block of x meets D, which happens for finitely many x. So f' is a finite modification of f, whose phase is resolving, and Proposition 3.7(c) gives $f' = f$, hence $Y' = Y$. □

Definition 3.11 (Walls)**.** Let $(\varsigma, \mathfrak{a}) \in \mathbb{A}^{\mathrm{q}}$, where $\mathfrak{a}$ has thick type κ. A thick type has a *vertical red atom* if its north or south endpoint is red, and a *horizontal red atom* if its east or west endpoint is red.

- The pair lies on a *vertical red path* if κ has a vertical red atom, $\varsigma_1 = 1$ if this atom is r_0 and $\varsigma_1 = 0$ if it is r_2, and, when κ is a corner, $\varsigma_2 = 1$ if its north endpoint is red and $\varsigma_2 = 0$ if its south endpoint is red. It is a *left side* if the atom is r_0 and a *right side* if it is r_2. If moreover $\chi_1(\mathfrak{a}) = 1$, the pair lies on a *light vertical wall*.

- The pair lies on a *horizontal red path* if κ has a horizontal red atom, $\varsigma_2 = 1$ if this atom is r_1 and $\varsigma_2 = 0$ if it is r_3, and, when κ is a corner, $\varsigma_1 = 1$ if its east endpoint is red and $\varsigma_1 = 0$ if its west endpoint is red. It is a *bottom side* if the atom is r_1 and a *top side* if it is r_3. If moreover $\chi_2(\mathfrak{a}) = 1$, the pair lies on a *light horizontal wall*.

On each axis, the red endpoints of a thick type carry at most one atom value (Table 2; a line carries its atom at both ends), so these notions are well defined. A corner quarter can lie on a light vertical and a light horizontal wall at once; a light vertical wall is not a light horizontal wall unless it is such a corner. The two notions are tested separately below.

3.5 Frames

Definition 3.12 (Frames, free coordinates and owners)**.** Let $\tau_i \in \mathbb{Z}_2$ and, for $d \geq 1$, let $a_d \in [0, 4^{d+1})$ with $a_d \equiv \tau_i \pmod{4^{d+1}}$. The *frames of depth d* are the integer intervals

$$I_d(m) = \left[a_d + (4m+1)4^d + 1,\ a_d + (4m+3)4^d\right], \qquad m \in \mathbb{Z},$$

with *left endpoint* $a_d + (4m+1)4^d + 1$ and *right endpoint* $a_d + (4m+3)4^d$. The *free coordinates* of $I_d(m)$ are the elements of

$$\mathcal{F}_d(m) = \left\{z \in \mathring{I}_d(m) :\ z \notin I_e(m') \text{ for all } 1 \leq e < d,\ m' \in \mathbb{Z}\right\},$$

where $\mathring{I}$ is I without its endpoints. If z lies in some frame, its *owner depth* $\mathrm{own}(z)$ is the least d such that z lies in a frame of depth d. For a quarter phase $\tau = (\tau_1, \tau_2)$ we write $I_d^{(i)}(m)$, $\mathcal{F}_d^{(i)}(m)$ and own_i for the objects defined from τ_i, and call $I^{(1)}$ the *column frames* and $I^{(2)}$ the *row frames*.

Lemma 3.13 (Frames)**.** *Fix $\tau_i \in \mathbb{Z}_2$ and $d \geq 1$, and put $N_d = 2^{d+2} - 2$.*

(a) Frames of the same depth are disjoint. If $e < d$, the endpoints of every frame of depth d are congruent to a_e and $a_e + 1$ modulo 4^{e+1} and lie in no frame of depth e. Two frames of different depths are disjoint, or the one of smaller depth lies in the interior of the other.

(b) If $e < d$, every frame of depth d contains a frame of depth e.

(c) $|\mathcal{F}_d(m)| = N_d$. The least element of $\mathcal{F}_d(m)$ is $r_d(m) = a_d + (4m+1)4^d + 2$, the left endpoint plus one, and the largest is the right endpoint minus one.

(d) $r_d(m) \equiv a_d + 4^d + 2 \pmod{4^{d+1}}$, and for $D > d$ every $r_D(m')$ is congruent to $a_d + 2$ modulo 4^{d+1}. Hence no integer is of the form $r_d(m)$ for two different depths d.

(e) If the base-4 expansion of $z - \tau_i$ has the digit 2 in some position $d \geq 1$, then z lies in the interior of a frame of depth d.

Proof. (a), (b) and the count in (c) are proved in [12]; the proof is recalled in Section A. For the rest of (c), the left endpoint plus one is congruent to $a_e + 2$ modulo 4^{e+1} for every $e < d$, and an integer z lies in a frame of depth e exactly when $(z - a_e) \bmod 4^{e+1} \in [4^e + 1, 3 \cdot 4^e]$, which excludes 2; so it is free. The right endpoint minus one is congruent to $a_e - 1$, that is to $4^{e+1} - 1$ relative to a_e, which is also excluded. (d) The first congruence is immediate; for $D > d$, 4^{d+1} divides $(4m'+1)4^D$ and $a_D \equiv a_d \pmod{4^{d+1}}$. The two residues differ by $4^d \not\equiv 0$. (e) The digit condition says $(z - a_d) \bmod 4^{d+1} \in [2 \cdot 4^d, 3 \cdot 4^d - 1]$, and these residues lie in the interior $[4^d + 2, 3 \cdot 4^d - 1]$ of the frame window. □

Lemma 3.14 (Light walls). *Let Y be a tiling of $\mathbb{A}^{\mathfrak{q}}$ with quarter phase τ, where $\tau_1, \tau_2 \notin \mathbb{Z}$. Then $Y(\xi)$ lies on a light vertical wall if and only if there are $d \geq 1$ and $m, m' \in \mathbb{Z}$ such that ξ_1 is an endpoint of $I_d^{(1)}(m)$ and $\xi_2 \in I_d^{(2)}(m')$; it is a left side when ξ_1 is the left endpoint and a right side when ξ_1 is the right endpoint. Dually, $Y(\xi)$ lies on a light horizontal wall if and only if ξ_2 is an endpoint of some $I_d^{(2)}(m')$ and $\xi_1 \in I_d^{(1)}(m)$ for some m, with a bottom side at the left endpoint and a top side at the right endpoint.*

The proof, which uses the red paths of the tower, is in Section A. The condition $\tau_i \notin \mathbb{Z}$ ensures that every column and row of the parent tiling has a finite rank there.

Definition 3.15 (Typical phases). A quarter phase $\tau \in \mathbb{Z}_2^2$ is *typical* if it is resolving, $\tau_1, \tau_2 \notin \mathbb{Z}$, and for $i = 1, 2$ and every $z \in \mathbb{Z}$ the base-4 expansion of $z - \tau_i$ has the digit 2 in some position $d \geq 1$.

Haar-almost every τ is typical: the first condition was checked above, $\mathbb{Z}^2$ is countable, and for fixed z the base-4 digits of $z - \tau_i$ are independent and uniform. By Lemma 3.13(e), for a typical τ every integer has a finite owner depth in each direction.

4 The interaction

This section defines the interaction Φ of Theorem A completely. A site value is a tuple of fields; the interaction consists of a set $\mathcal{T}$ of allowed site values, equality of designated edge components between neighbours, and a charge in $\{0, 1\}$ on each site value. The set $\mathcal{T}$ is defined by the rules (O1)–(P3) and (C0)–(C9) below: a site value is allowed exactly when it satisfies all of them. Each rule is a Boolean condition on the fields of one site, so a reader can list $\mathcal{T}$ by checking the rules on the finite product of the value sets. The fields and colours are tabulated in Section B, and the machine M, with its local rule f_M, is given in Section C. Section 5 explains what the rules produce.

4.1 Fields

The machine M of Section C has a finite set $\mathcal{S}_M$ of *cell contents*: pairs of a tape symbol and either a state or the mark $\cdot$ for no head. It has a start state q_{start}, an accepting state q_{acc}, a local rule $f_M \colon (\mathcal{S}_M \cup \{\perp\}) \times \mathcal{S}_M \times (\mathcal{S}_M \cup \{\perp\}) \to \mathcal{S}_M$, an initialization map $\text{init} \colon \{0,1\}^2 \to \mathcal{S}_M$, and two *mask bits* $\mathsf{m}^-(s), \mathsf{m}^+(s) \in \{0, 1\}$ of each $s \in \mathcal{S}_M$. Let $\circ$ be a new symbol, the *empty colour*.

A site value ω_ξ consists of the following fields.

(F1) A *quarter* $\mathfrak{q} \in \mathbb{A}^{\mathfrak{q}}$ (Definition 3.8).

(F2) Four *orientation tags* $o^{\mathrm{W}}, o^{\mathrm{E}} \in \{\mathrm{L}, \mathrm{R}\}^2$ and $o^{\mathrm{S}}, o^{\mathrm{N}} \in \{\mathrm{B}, \mathrm{T}\}^2$. A tag is a pair $o = (o_-, o_+)$.

(F3) Four *event tags* $e^{\mathrm{W}}, e^{\mathrm{E}}, e^{\mathrm{S}}, e^{\mathrm{N}} \in \{\text{cell}, \text{wall}\}^2 \cup \{*\}$, a pair $e = (e_-, e_+)$ or the symbol $*$.

(F4) Four *logical colours* $\ell^{\mathrm{W}}, \ell^{\mathrm{E}} \in \Lambda_{\mathrm{h}} = \{\circ\} \cup \Lambda_{\mathrm{h}}^+$ and $\ell^{\mathrm{S}}, \ell^{\mathrm{N}} \in \Lambda_{\mathrm{v}} = \{\circ\} \cup \Lambda_{\mathrm{v}}^+$, where

$$\Lambda_{\mathrm{h}}^+ = \underbrace{\{0,1\}}_{\mathrm{ct}} \times \underbrace{\mathcal{S}_M}_{\mathrm{tb}} \times \underbrace{\{0,1\}}_{\mathrm{hot}} \times \underbrace{\{0,1\}}_{\mathrm{rdy}} \times \underbrace{\{0,1\}^2}_{g^-, g^+},$$

$$\Lambda_{\mathrm{v}}^+ = \underbrace{\{0,1\}}_{\mathrm{cy}} \times \underbrace{\{0,1\}}_{\mathrm{ay}} \times \underbrace{[0,80]}_{\mathrm{rc}} \times \underbrace{\{0,1\}}_{\mathrm{rdy}} \times \underbrace{\mathcal{S}_M^2}_{\mathrm{tb}} \times \underbrace{\{0,1\}^4}_{x}.$$

We write $\ell^X_{\mathrm{ct}}, \ell^X_{\mathrm{tb}}, \ldots$ for the components of $\ell^X \neq \circ$.

(F5) A *cell record* $\mathfrak{k} \in \mathcal{K} \cup \{\circ\}$, where an element of $\mathcal{K}$ has the components

$$\mathrm{ct}_{\mathrm{in}}, \mathrm{ct}, \mathrm{cy}, \mathrm{ay}, \mathrm{sm}, \mathrm{hot}, \mathrm{rdy}, g^-, g^+ \in \{0,1\}, \quad \mathrm{rc} \in [0,80], \quad t_{\mathrm{S}}, t_{\mathrm{N}} \in \mathcal{S}_M \cup \{\perp\},\ t_{\mathrm{C}} \in \mathcal{S}_M,$$

$$x = (x_{\mathrm{h}}^-, x_{\mathrm{h}}^+, x_{\mathrm{v}}^-, x_{\mathrm{v}}^+) \in \{0,1\}^4, \qquad y = (y_{\mathrm{h}}^-, y_{\mathrm{h}}^+, y_{\mathrm{v}}^-, y_{\mathrm{v}}^+) \in \{0,1\}^4.$$

(F6) A *row marker* $\mathfrak{r} \in \{0,1\}$ and a *column marker* $\mathfrak{c} \in \{0,1\}$.

(F7) A *spin* $\sigma \in \{+1,-1\}$.

(F8) Four *wire values* $w^{\mathrm{W}}, w^{\mathrm{E}}, w^{\mathrm{S}}, w^{\mathrm{N}} \in \{+1,-1,\square\}$, where $\square$ is blank.

The site alphabet $\mathfrak{S}$ is the set of all such tuples, the product of the value sets listed.

4.2 Colours and the interaction

The four *colours* of a site value are

$$c_{\mathrm{E}}(\omega_\xi) = (\mathfrak{q}_{\mathrm{E}}, o^{\mathrm{E}}, e^{\mathrm{E}}, \ell^{\mathrm{E}}, \mathfrak{r}, w^{\mathrm{E}}), \qquad c_{\mathrm{W}}(\omega_\xi) = (\mathfrak{q}_{\mathrm{W}}, o^{\mathrm{W}}, e^{\mathrm{W}}, \ell^{\mathrm{W}}, \mathfrak{r}, w^{\mathrm{W}}),$$
$$c_{\mathrm{N}}(\omega_\xi) = (\mathfrak{q}_{\mathrm{N}}, o^{\mathrm{N}}, e^{\mathrm{N}}, \ell^{\mathrm{N}}, \mathfrak{c}, w^{\mathrm{N}}), \qquad c_{\mathrm{S}}(\omega_\xi) = (\mathfrak{q}_{\mathrm{S}}, o^{\mathrm{S}}, e^{\mathrm{S}}, \ell^{\mathrm{S}}, \mathfrak{c}, w^{\mathrm{S}}),$$

where $\mathfrak{q}_X$ is the colour of $\mathfrak{q}$ in $\mathbb{A}^{\mathfrak{q}}$.

Definition 4.1 (The interaction)**.** Let $\mathcal{T} \subseteq \mathfrak{S}$ be the set of site values satisfying (O1)–(P3) and (C0)–(C9), and let $\mathrm{ch}\colon \mathfrak{S} \to \{0,1\}$ be the charge of Section 4.6. Put $\Omega = \mathfrak{S}^{\mathbb{Z}^2}$ and

$$\Phi_{\{\xi\}}(\omega) = \begin{cases} \mathrm{ch}(\omega_\xi) & \omega_\xi \in \mathcal{T}, \\ +\infty & \omega_\xi \notin \mathcal{T}, \end{cases} \qquad \Phi_{\{\xi,\xi+\mathbf{e}_1\}}(\omega) = \begin{cases} 0 & c_{\mathrm{E}}(\omega_\xi) = c_{\mathrm{W}}(\omega_{\xi+\mathbf{e}_1}), \\ +\infty & \text{otherwise}, \end{cases}$$

$\Phi_{\{\xi,\xi+\mathbf{e}_2\}}(\omega) = 0$ if $c_{\mathrm{N}}(\omega_\xi) = c_{\mathrm{S}}(\omega_{\xi+\mathbf{e}_2})$ and $+\infty$ otherwise, and $\Phi_A = 0$ for all other A.

Restricting the alphabet to $\mathcal{T}$ gives an equivalent model; the single-site term is written out to show where the hard constraints sit. The charge is defined on all of $\mathfrak{S}$ by the formula of Section 4.6, read as 0 when the cell record is $\circ$. Φ is nearest-neighbour and translation invariant, and its values lie in $\{0,1,+\infty\}$. A configuration is legal exactly when every site value lies in $\mathcal{T}$ and neighbouring colours agree. Matching colours makes the quarters a tiling of $\mathbb{A}^{\mathfrak{q}}$, makes $\mathfrak{r}$ constant along rows and $\mathfrak{c}$ constant along columns, and identifies the tags, logical colours and wire values on the two sides of each edge.

4.3 Predicates read from a site value

The following quantities are functions of one site value.

- $\mathrm{VW}(\mathfrak{q}) \in \{-, \mathrm{L}, \mathrm{R}\}$: L if $\mathfrak{q}$ lies on a light vertical wall as a left side, R if as a right side, $-$ otherwise. $\mathrm{HW}(\mathfrak{q}) \in \{-, \mathrm{B}, \mathrm{T}\}$: likewise for light horizontal walls, with bottom and top sides (Definition 3.11).
- The site is a *horizontal carrier* if $\mathrm{VW} = -$ and $o^{\mathrm{W}} = (\mathrm{L}, \mathrm{R})$, and a *vertical carrier* if $\mathrm{HW} = -$ and $o^{\mathrm{S}} = (\mathrm{B}, \mathrm{T})$. It is *active* if it is both, *horizontal* if it is only a horizontal carrier, *vertical* if it is only a vertical carrier, and *inert* if it is neither.
- At an active site, the *boundary flags* are $f_{\mathrm{W}} = [e_-^{\mathrm{W}} = \text{wall}]$, $f_{\mathrm{E}} = [e_+^{\mathrm{E}} = \text{wall}]$, $f_{\mathrm{S}} = [e_-^{\mathrm{S}} = \text{wall}]$ and $f_{\mathrm{N}} = [e_+^{\mathrm{N}} = \text{wall}]$, where $[\cdot]$ is 1 for true and 0 for false. An active site with $f_{\mathrm{W}} = f_{\mathrm{S}} = 1$ is a *root*.

4.4 Routing, markers and wires

In a tag $o = (o_-, o_+)$ on a horizontal edge, o_- is meant to be the side of the nearest light vertical wall to the west of the edge and o_+ that of the nearest one to the east; for vertical edges, south and north. In an event tag, e_- and e_+ record whether the nearest event on that side is an active site (cell) or a wall.

(O1) If $\mathrm{VW} = -$, then $o^{\mathrm{W}} = o^{\mathrm{E}}$. If $\mathrm{VW} = \mathrm{L}$, then $o^{\mathrm{W}}_+ = \mathrm{L}$ and $o^{\mathrm{E}}_- = \mathrm{L}$. If $\mathrm{VW} = \mathrm{R}$, then $o^{\mathrm{W}}_+ = \mathrm{R}$ and $o^{\mathrm{E}}_- = \mathrm{R}$.

(O2) If $\mathrm{HW} = -$, then $o^{\mathrm{S}} = o^{\mathrm{N}}$. If $\mathrm{HW} = \mathrm{B}$, then $o^{\mathrm{S}}_+ = \mathrm{B}$ and $o^{\mathrm{N}}_- = \mathrm{B}$. If $\mathrm{HW} = \mathrm{T}$, then $o^{\mathrm{S}}_+ = \mathrm{T}$ and $o^{\mathrm{N}}_- = \mathrm{T}$.

(E1) $e^X = *$ if and only if $o^X \neq (\mathrm{L}, \mathrm{R})$, for $X \in \{\mathrm{W}, \mathrm{E}\}$; and $e^X = *$ if and only if $o^X \neq (\mathrm{B}, \mathrm{T})$, for $X \in \{\mathrm{S}, \mathrm{N}\}$.

(E2) At a horizontal site, $e^{\mathrm{W}} = e^{\mathrm{E}}$. At an active site, $e^{\mathrm{W}}_+ =$ cell and $e^{\mathrm{E}}_- =$ cell. If $\mathrm{VW} = \mathrm{L}$ and $o^{\mathrm{E}} = (\mathrm{L}, \mathrm{R})$, then $e^{\mathrm{E}}_- =$ wall. If $\mathrm{VW} = \mathrm{R}$ and $o^{\mathrm{W}} = (\mathrm{L}, \mathrm{R})$, then $e^{\mathrm{W}}_+ =$ wall.

(E3) At a vertical site, $e^{\mathrm{S}} = e^{\mathrm{N}}$. At an active site, $e^{\mathrm{S}}_+ =$ cell and $e^{\mathrm{N}}_- =$ cell. If $\mathrm{HW} = \mathrm{B}$ and $o^{\mathrm{N}} = (\mathrm{B}, \mathrm{T})$, then $e^{\mathrm{N}}_- =$ wall. If $\mathrm{HW} = \mathrm{T}$ and $o^{\mathrm{S}} = (\mathrm{B}, \mathrm{T})$, then $e^{\mathrm{S}}_+ =$ wall.

(K1) At a site that is not a horizontal carrier, $\ell^{\mathrm{W}} = \ell^{\mathrm{E}} = \circ$. At a horizontal site, $\ell^{\mathrm{W}} = \ell^{\mathrm{E}}$.

(K2) At a site that is not a vertical carrier, $\ell^{\mathrm{S}} = \ell^{\mathrm{N}} = \circ$. At a vertical site, $\ell^{\mathrm{S}} = \ell^{\mathrm{N}}$.

(K3) At a site that is not active, $\mathfrak{k} = \circ$. At an active site, $\mathfrak{k} \in \mathcal{K}$ and (C0)–(C9) of Section 4.5 hold.

(Q1) At an active site, $\mathfrak{r} = f_{\mathrm{S}}$ and $\mathfrak{c} = f_{\mathrm{W}}$.

(Q2) If $\mathrm{HW} \neq -$, then $\mathfrak{r} = 0$. If $\mathrm{VW} \neq -$, then $\mathfrak{c} = 0$.

(P1) If $\mathfrak{r} = 0$, then $w^{\mathrm{W}} = w^{\mathrm{E}} = \square$. If $\mathfrak{r} = 1$, then $w^{\mathrm{W}}, w^{\mathrm{E}} \in \{+1, -1\}$, and $w^{\mathrm{E}} = \sigma$ at a root and $w^{\mathrm{E}} = w^{\mathrm{W}}$ at every other site.

(P2) If $\mathfrak{c} = 0$, then $w^{\mathrm{S}} = w^{\mathrm{N}} = \square$. If $\mathfrak{c} = 1$, then $w^{\mathrm{S}}, w^{\mathrm{N}} \in \{+1, -1\}$, and $w^{\mathrm{N}} = \sigma$ at a root and $w^{\mathrm{N}} = w^{\mathrm{S}}$ at every other site.

(P3) At a root with $\mathrm{rdy} = 1$: if $(x^-_{\mathrm{h}}, x^+_{\mathrm{h}}) = (0, 1)$ then $\sigma = w^{\mathrm{W}}$; if $(x^-_{\mathrm{v}}, x^+_{\mathrm{v}}) = (0, 1)$ then $\sigma = w^{\mathrm{S}}$.

Rule (Q2) seeds the row marker with 0 only on light *horizontal* walls, and the column marker only on light *vertical* walls. The orientation matters: a root row crosses light vertical walls (Lemma 5.6), so a seed testing every light wall would admit no legal configuration with a typical scaffold.

4.5 Cells

The rules (C0)–(C9) apply at active sites, where $\mathfrak{k} \in \mathcal{K}$. A statement about a component of ℓ^X is required only when $\ell^X \neq \circ$; by (C0) this is when the corresponding boundary flag is 0. Write $\oplus$ for addition modulo 2, $\wedge$ for the product of bits, and maj for the majority of three bits.

(C0) *Boundaries.* $\ell^{\mathrm{W}} = \circ$ if and only if $f_{\mathrm{W}} = 1$; $\ell^{\mathrm{E}} = \circ$ if and only if $f_{\mathrm{E}} = 1$; $\ell^{\mathrm{S}} = \circ$ if and only if $f_{\mathrm{S}} = 1$; $\ell^{\mathrm{N}} = \circ$ if and only if $f_{\mathrm{N}} = 1$.

(C1) *Counter.* If $f_{\mathrm{W}} = 1$: $\mathrm{ct}_{\mathrm{in}} = \mathrm{ct} = \mathrm{cy} = 0$. If $f_{\mathrm{W}} = 0$: $\mathrm{ct}_{\mathrm{in}} = \ell^{\mathrm{W}}_{\mathrm{ct}}$; $\mathrm{cy} = 1$ if $f_{\mathrm{S}} = 1$ and $\mathrm{cy} = \ell^{\mathrm{S}}_{\mathrm{cy}}$ if $f_{\mathrm{S}} = 0$; $\mathrm{ct} = \mathrm{ct}_{\mathrm{in}} \oplus \mathrm{cy}$. In all cases $\ell^{\mathrm{E}}_{\mathrm{ct}} = \mathrm{ct}$ and $\ell^{\mathrm{N}}_{\mathrm{cy}} = \mathrm{ct}_{\mathrm{in}} \wedge \mathrm{cy}$.

(C2) *Row counter.* $\mathrm{rc} = 0$ if $f_{\mathrm{S}} = 1$, and $\mathrm{rc} = \min(\ell^{\mathrm{S}}_{\mathrm{rc}} + 1, 80)$ if $f_{\mathrm{S}} = 0$. $\ell^{\mathrm{N}}_{\mathrm{rc}} = \mathrm{rc}$.

(C3) *Adder.* If $f_{\mathrm{E}} = 0$: $\mathrm{ay} = \mathrm{sm} = 0$ and $\ell^{\mathrm{N}}_{\mathrm{ay}} = 0$. If $f_{\mathrm{E}} = 1$: with $k = [\mathrm{rc} \le 1]$, $\mathrm{ay} = 0$ if $f_{\mathrm{S}} = 1$ and $\mathrm{ay} = \ell^{\mathrm{S}}_{\mathrm{ay}}$ if $f_{\mathrm{S}} = 0$; $\mathrm{sm} = \mathrm{ct} \oplus k \oplus \mathrm{ay}$; $\ell^{\mathrm{N}}_{\mathrm{ay}} = \mathrm{maj}(\mathrm{ct}, k, \mathrm{ay})$.

(C4) *One-hot marker.* $\mathrm{hot} = \mathrm{sm}$ if $f_{\mathrm{E}} = 1$, and $\mathrm{hot} = \ell^{\mathrm{E}}_{\mathrm{hot}}$ if $f_{\mathrm{E}} = 0$. $\ell^{\mathrm{W}}_{\mathrm{hot}} = \mathrm{hot}$.

(C5) *Tableau of M.* $t_{\mathrm{S}} = \perp$ if $f_{\mathrm{S}} = 1$, and $\ell^{\mathrm{S}}_{\mathrm{tb}} = (t_{\mathrm{S}}, t_{\mathrm{C}})$ if $f_{\mathrm{S}} = 0$. $t_{\mathrm{N}} = \perp$ if $f_{\mathrm{N}} = 1$, and $\ell^{\mathrm{N}}_{\mathrm{tb}} = (t_{\mathrm{C}}, t_{\mathrm{N}})$ if $f_{\mathrm{N}} = 0$. $t_{\mathrm{C}} = \mathrm{init}(\mathrm{hot}, f_{\mathrm{S}})$ if $f_{\mathrm{W}} = 1$, and $t_{\mathrm{C}} = \ell^{\mathrm{W}}_{\mathrm{tb}}$ if $f_{\mathrm{W}} = 0$. $\ell^{\mathrm{E}}_{\mathrm{tb}} = f_M(t_{\mathrm{S}}, t_{\mathrm{C}}, t_{\mathrm{N}})$.

(C6) *Readiness.* If $f_{\mathrm{E}} = 1$: $\mathrm{rdy} = [t_{\mathrm{C}}$ carries the state $q_{\mathrm{acc}}]$ if $f_{\mathrm{S}} = 1$, $\mathrm{rdy} = \ell^{\mathrm{S}}_{\mathrm{rdy}}$ if $f_{\mathrm{S}} = 0$, and $\ell^{\mathrm{N}}_{\mathrm{rdy}} = \mathrm{rdy}$. If $f_{\mathrm{E}} = 0$: $\mathrm{rdy} = \ell^{\mathrm{E}}_{\mathrm{rdy}}$ and $\ell^{\mathrm{N}}_{\mathrm{rdy}} = 0$. In all cases $\ell^{\mathrm{W}}_{\mathrm{rdy}} = \mathrm{rdy}$.

(C7) *Masks.* If $f_{\mathrm{E}} = 1$: $g^{\pm} = \mathrm{rdy} \wedge \mathsf{m}^{\pm}(t_{\mathrm{C}})$. If $f_{\mathrm{E}} = 0$: $g^{\pm} = \ell^{\mathrm{E}}_{g^{\pm}}$. In all cases $\ell^{\mathrm{W}}_{g^{\pm}} = g^{\pm}$.

(C8) *Switches*, for each of the four components x_ι of x separately. If $f_{\mathrm{W}} = 0$: $x_\iota = 0$ and $\ell^{\mathrm{N}}_{x_\iota} = 0$. If $f_{\mathrm{W}} = 1$ and $f_{\mathrm{S}} = 1$: $x_\iota \in \{0, 1\}$ is unconstrained if $\mathrm{rdy} = 1$, and $x_\iota = 0$ if $\mathrm{rdy} = 0$. If $f_{\mathrm{W}} = 1$ and $f_{\mathrm{S}} = 0$: $x_\iota = \ell^{\mathrm{S}}_{x_\iota}$. If $f_{\mathrm{W}} = 1$: $\ell^{\mathrm{N}}_{x_\iota} = x_\iota$.

(C9) *Free bits.* If $f_{\mathrm{W}} = 0$: $y = 0$. If $f_{\mathrm{W}} = 1$: for $k \in \{\mathrm{h}, \mathrm{v}\}$ and each sign, $y^{\pm}_k \in \{0, 1\}$ is unconstrained if $x^{\pm}_k = g^{\pm} = 1$, and $y^{\pm}_k = 0$ otherwise.

4.6 The charge

Order the switches as $(x_0, x_1, x_2, x_3) = (x^-_{\mathrm{h}}, x^+_{\mathrm{h}}, x^-_{\mathrm{v}}, x^+_{\mathrm{v}})$. The charge of a site value is

$$\mathrm{ch}(\omega_\xi) = \begin{cases} x_{\lfloor \mathrm{rc}/20 \rfloor} & \text{if the site is active, } f_{\mathrm{W}} = 1 \text{ and } \mathrm{rc} \le 79, \\ 0 & \text{otherwise.} \end{cases}$$

These are the only finite nonzero energies. At each site at most one switch is charged, so the energy of a site value is 0, 1 or $+\infty$.

4.7 Background and payload

The *payload* coordinates of a site are its spin σ, its four wire values, the components x and y of its cell record, and the components x of ℓ^{S} and ℓ^{N}. All other fields and components form the *background* of the site. A background configuration is a map from $\mathbb{Z}^2$ to background values; for a configuration ω we write $\omega = (\mathfrak{b}, p)$ with background $\mathfrak{b}$ and payload p.

Only the rules (P1)–(P3), (C8), (C9) and the charge involve payload coordinates, and each of them is a conjunction of conditions of the following kinds: on the wire values, σ and background fields; on one switch component and background fields; on one free bit, the switch of the same name and background fields; and, in (P3), on σ, one incoming wire value, the two switches of one sensor and background fields. This splitting is used in Lemma 6.1.

5 The background

In this section ω is a legal configuration whose quarters form a tiling Y of $\mathbb{A}^{\mathsf{q}}$ with typical quarter phase τ (Definition 3.15). We show that the rules force every background field of ω, compute the result, and deduce that the background has one translation-invariant law and cannot be changed by finite modifications. Frames, free coordinates and owners refer to τ: column frames $I^{(1)}_d(m)$ and row frames $I^{(2)}_d(m)$, with $r^{(i)}_d(m) = \min \mathcal{F}^{(i)}_d(m)$.

5.1 Corridors, carriers and grids

Fix a row $y \in \mathbb{Z}$. Let $D(y)$ be the set of depths d such that y lies in a row frame of depth d, so that $\mathrm{own}_2(y) = \min D(y)$, and let $\mathcal{J}(y)$ be the set of column frames of the depths in $D(y)$. By Lemma 3.14, the sites of row y that lie on light vertical walls are exactly the sites (z, y) with z an endpoint of a member of $\mathcal{J}(y)$, with left sides at left endpoints and right sides at right endpoints. We call these sites the *walls* of the row. By Lemma 3.13(a), $\mathcal{J}(y)$ is laminar in the sense of Lemma A.8, and its members that contain no other member are exactly the frames of depth $\mathrm{own}_2(y)$: a frame of larger depth contains one of depth $\mathrm{own}_2(y)$ by Lemma 3.13(b), and a member inside a frame of depth $\mathrm{own}_2(y)$ would have smaller depth. The frames of depth $\mathrm{own}_2(y)$ occur periodically in both directions, so the walls of the row are unbounded on both sides. Columns are treated in the same way with own_1, row frames and light horizontal walls.

Proposition 5.1 (Routing). *(a) Let $\xi \in \mathbb{Z}^2$, let $w_- \le \xi_1$ be the largest wall of row ξ_2 at or west of ξ, and let $w_+ \ge \xi_1 + 1$ be the smallest wall at or east of $\xi + \mathbf{e}_1$. Then $o^{\mathrm{E}}(\xi) = o^{\mathrm{W}}(\xi + \mathbf{e}_1)$ is the pair of the sides of w_- and w_+. The vertical tags are described in the same way.*

(b) ξ is a horizontal carrier if and only if ξ_1 lies in the interior of a column frame of depth $\mathrm{own}_2(\xi_2)$, and a vertical carrier if and only if ξ_2 lies in the interior of a row frame of depth $\mathrm{own}_1(\xi_1)$.

(c) ξ is active if and only if $\xi \in \mathcal{F}_d^{(1)}(m_1) \times \mathcal{F}_d^{(2)}(m_2)$ for some $d \ge 1$ and $m_1, m_2 \in \mathbb{Z}$.

Proof. (a) Let $w < w'$ be consecutive walls of row ξ_2. At each site strictly between them $\mathrm{VW} = -$, so by (O1) its west and east tags are equal, and by matching all edges between w and w' carry one tag. By (O1) at w its first entry is the side of w, and by (O1) at w' its second entry is the side of w'. Every edge of the row lies between two consecutive walls.

(b) If ξ is a wall, it is not a horizontal carrier, and ξ_1 lies in the interior of no frame of depth $\mathrm{own}_2(\xi_2)$, whose endpoints are consecutive walls by Lemma A.8. Otherwise let $w_- < \xi_1 < w_+$ be the nearest walls. By (a), ξ is a horizontal carrier if and only if w_- is a left side and w_+ a right side. By Lemma A.8 this holds if and only if $[w_-, w_+]$ is a member of $\mathcal{J}(\xi_2)$ with consecutive endpoints, that is a member containing no other member, that is a frame of depth $\mathrm{own}_2(\xi_2)$. Columns are the same.

(c) If ξ is active, then ξ_1 lies in a column frame of depth $\mathrm{own}_2(\xi_2)$, so $\mathrm{own}_1(\xi_1) \le \mathrm{own}_2(\xi_2)$, and symmetrically. With d the common value, ξ_1 lies in the interior of a column frame of depth d and in no column frame of smaller depth, that is $\xi_1 \in \mathcal{F}_d^{(1)}(m_1)$; likewise $\xi_2 \in \mathcal{F}_d^{(2)}(m_2)$. Conversely, if $\xi_i \in \mathcal{F}_d^{(i)}(m_i)$, then $\mathrm{own}_i(\xi_i) = d$ for both i, and (b) shows that ξ is a horizontal and a vertical carrier. □

Horizontal walls make every frame endpoint row a row of walls; its corridors carry the tag (L, R) where they cross frames of the owner depth, but contain no active site.

Definition 5.2 (Grids). For $d \ge 1$ and $m_1, m_2 \in \mathbb{Z}$ put $N = N_d = 2^{d+2} - 2$ and let $\zeta_0^{(i)} < \cdots < \zeta_{N-1}^{(i)}$ be the elements of $\mathcal{F}_d^{(i)}(m_i)$ (Lemma 3.13(c)). The *grid* $G_d(m_1, m_2)$ is the set of sites $\xi(i,j) = (\zeta_i^{(1)}, \zeta_j^{(2)})$, $i, j \in [0, N)$; the pair (i, j) is the *logical position* of $\xi(i,j)$. Its *root* is $\mathbf{r}_d(m_1, m_2) = \xi(0,0) = (r_d^{(1)}(m_1), r_d^{(2)}(m_2))$. The *label* of depth d is $n(d) = 1 + v_2(d)$, where $v_2(d)$ is the exponent of 2 in d.

Lemma 5.3 (Grids). *Let $G = G_d(m_1, m_2)$.*

(a) The sites strictly between $\xi(i,j)$ and $\xi(i+1,j)$ are horizontal, and those strictly between $\xi(i,j)$ and $\xi(i,j+1)$ are vertical.

(b) $\xi(0,j)-\mathbf{e}_1$ *and* $\xi(N-1,j)+\mathbf{e}_1$ *lie on light vertical walls, as a left and a right side;* $\xi(i,0)-\mathbf{e}_2$ *and* $\xi(i,N-1)+\mathbf{e}_2$ *lie on light horizontal walls, as a bottom and a top side.*

(c) The boundary flags at $\xi(i,j)$ *are* $f_{\mathrm{W}}=[i=0]$, $f_{\mathrm{E}}=[i=N-1]$, $f_{\mathrm{S}}=[j=0]$ *and* $f_{\mathrm{N}}=[j=N-1]$.

(d) Every active site lies in exactly one grid, and the roots of ω *are exactly the roots of the grids.*

Proof. (a) Let $\zeta_i^{(1)}<z<\zeta_{i+1}^{(1)}$. Then z lies in the interior of $I_d^{(1)}(m_1)$ but is not free, so it lies in a column frame of depth less than d, and $\mathrm{own}_1(z)<d$. Since $\mathrm{own}_2(\zeta_j^{(2)})=d$, Proposition 5.1(b) shows that $(z,\zeta_j^{(2)})$ is a horizontal carrier, and it is not a vertical carrier, because $\zeta_j^{(2)}$ lies in no row frame of depth $\mathrm{own}_1(z)<d$. The vertical case is the same.

(b) By Lemma 3.13(c), $\zeta_0^{(1)}-1$ and $\zeta_{N-1}^{(1)}+1$ are the endpoints of $I_d^{(1)}(m_1)$, and $\zeta_j^{(2)}\in I_d^{(2)}(m_2)$; apply Lemma 3.14.

(c) The frame $I_d^{(1)}(m_1)$ contains no other member of $\mathcal{J}(\zeta_j^{(2)})$, so its endpoints are consecutive walls of the row, and the sites between them are the N active sites $\xi(i,j)$ and the horizontal sites of (a). On the edges between these walls the orientation tag is (L,R), so the event tags are pairs by (E1). Reading (E2) from west to east: the left wall sets $e_-=$ wall on its east edge, horizontal sites copy, and each active site sets $e_-=$ cell on its east edge. So the first entry of $e^{\mathrm{W}}(\xi(i,j))$ is wall exactly when $i=0$. The other three flags are the same.

(d) The first claim follows from Proposition 5.1(c), since the sets $\mathcal{F}_d^{(i)}(m)$ are pairwise disjoint for different (d,m) by Lemma 3.13(a). By (c), an active site is a root exactly when its logical position is $(0,0)$. □

5.2 The computation on a grid

Section C defines, for every $d\geq 1$, the *run* of M on the grid of depth d: tape columns $c_0,\dots,c_{N-1}\colon[0,N)\to\mathcal{S}_M$ with $c_0(j)=\mathrm{init}([j=d+2],[j=0])$ and $c_{i+1}(j)=f_M(c_i(j-1),c_i(j),c_i(j+1))$, where $c_i(-1)=c_i(N)=\perp$ (Definition C.1). The depth d is *ready* if $c_{N-1}(0)$ carries the state q_{acc}; write $\rho_d\in\{0,1\}$ for the indicator.

Lemma 5.4 (The grid computation)**.** *Let* $G=G_d(m_1,m_2)$ *and write* $\mathfrak{k}(i,j)$ *for the cell record at* $\xi(i,j)$*. The background components of the cell records of* G*, and of the logical colours at the sites of* G *and of its channels, are determined by* d*. In particular, for all* $i,j\in[0,N)$*:*

(a) $\mathrm{ct}(i,j)$ *is the binary digit of* i *at position* j*, and* $\mathrm{rc}(i,j)=\min(j,80)$*;*

(b) $\mathrm{hot}(i,j)=[j=d+2]$*;*

(c) $t_{\mathrm{C}}(i,j)=c_i(j)$*;*

(d) $\mathrm{rdy}(i,j)=\rho_d$ *and* $g^{\pm}(i,j)=\rho_d\,\mathsf{m}^{\pm}(c_{N-1}(j))$.

Proof. By Lemma 5.3(a) and (K1), (K2), the sites of a channel copy logical colours, so by matching the east colour of $\xi(i,j)$ is the west colour of $\xi(i+1,j)$ and the north colour of $\xi(i,j)$ is the south colour of $\xi(i,j+1)$. By Lemma 5.3(b),(c) and (K1), (K2), the colours facing the walls are $\circ$, in agreement with (C0). The rules (C1)–(C7) are then equations on the $N\times N$ logical grid, which we solve in order.

(a) Column 0 has $\mathrm{ct}=0$ by (C1). For $i\geq 1$, (C1) sets the carry into row 0 to 1, $\mathrm{ct}(i,j)=\mathrm{ct}(i-1,j)\oplus\mathrm{cy}(i,j)$ and $\mathrm{cy}(i,j+1)=\mathrm{ct}(i-1,j)\wedge\mathrm{cy}(i,j)$: column i is column $i-1$ plus one in binary, with the carry out of the top row dropped. Since $i<N<2^N$, column i holds i. The row counter is (C2).

(b) In the last column, $\mathrm{ct}(N-1,\cdot)$ holds $N-1 = 2^{d+2}-3$. By (C3) the adder adds the number with binary digits $k = [\mathrm{rc} \le 1] = [j \le 1]$, which is 3, with the carry chain running north from row 0, and drops the carry out of the top row. So $\mathrm{sm}(N-1,j)$ is the digit of 2^{d+2} at position j, which is $[j = d+2]$; this position exists because $d+2 < N$. By (C4) this value is copied west along each logical row.

(c) By (C5) and (b), column 0 holds $t_{\mathrm{C}}(0,j) = \mathrm{init}([j = d+2],[j=0]) = c_0(j)$. The triples $(t_{\mathrm{S}}, t_{\mathrm{C}}, t_{\mathrm{N}})$ at (i,j) are $(t_{\mathrm{C}}(i,j-1), t_{\mathrm{C}}(i,j), t_{\mathrm{C}}(i,j+1))$ with $\perp$ beyond the first and last rows, because the north colour $(t_{\mathrm{C}}(i,j), t_{\mathrm{N}}(i,j))$ of (i,j) equals the south colour $(t_{\mathrm{S}}(i,j+1), t_{\mathrm{C}}(i,j+1))$ of $(i,j+1)$. So $t_{\mathrm{C}}(i+1,j) = f_M(t_{\mathrm{C}}(i,j-1), t_{\mathrm{C}}(i,j), t_{\mathrm{C}}(i,j+1))$ by (C5), which is the recursion defining the run.

(d) By (C6), $\mathrm{rdy}(N-1,0) = \rho_d$, it is copied north along the last column and then west along each row. (C7) is the same for the masks.

The components $\mathrm{ct}_{\mathrm{in}}$, cy, ay and the colour components are determined by these values through (C1)–(C7). □

Lemma 5.5 (Readiness). *(a) If $\rho_d = 1$, then $d \ge 8$, U halts on $n(d)$, and $\mathsf{m}^-(c_{N-1}(j)) = [j < 20n(d) - 5]$ and $\mathsf{m}^+(c_{N-1}(j)) = [j < 20n(d) + 5]$ for all $j \in [0,N)$; moreover $20n(d) + 5 \le N_d$.*

(b) If U halts on n, there is d_0 such that $\rho_d = 1$ for every $d \ge d_0$ with $n(d) = n$; there are infinitely many such d.

The proof is in Section C. The machine M never reads the readiness bit or the masks of the cell records: its run depends only on d.

5.3 Markers and roots

Lemma 5.6 (Markers and roots). *(a) $\mathfrak{r}(\xi) = 1$ if and only if $\xi_2 = r_d^{(2)}(m)$ for some $d \ge 1$ and $m \in \mathbb{Z}$, and $\mathfrak{c}(\xi) = 1$ if and only if $\xi_1 = r_d^{(1)}(m)$ for some d and m. We call these rows and columns* root rows *and* root columns *of depth d; by Lemma 3.13(d) the depth is unique.*

(b) The roots on a root row of depth d are the sites $\mathbf{r}_d(m_1,m_2)$, $m_1 \in \mathbb{Z}$, for the fixed m_2; consecutive ones are 4^{d+1} apart. Root columns are the same.

(c) A root row of depth d contains light vertical walls, at the endpoints of the column frames of depth d, and no light horizontal wall.

Proof. Fix a row y and let $d = \mathrm{own}_2(y)$, which is finite because τ is typical.

Suppose first that y is an endpoint of a row frame $I_d^{(2)}(m')$. By Lemma 3.14 every site (z,y) with $z \in I_d^{(1)}(m)$ lies on a light horizontal wall, so $\mathfrak{r} = 0$ on the row by (Q2). The row contains no active site: an active site (z,y) would have $y \in \mathcal{F}_e^{(2)}(\cdot)$ for some e, so $e = \mathrm{own}_2(y) = d$ and y would be interior to a frame of depth d, while it is an endpoint of one, and frames of one depth are disjoint.

Otherwise y lies in the interior of $I_d^{(2)}(m')$ and in no row frame of smaller depth, so $y \in \mathcal{F}_d^{(2)}(m')$. By Proposition 5.1(c) the sites (z,y) with $z \in \mathcal{F}_d^{(1)}(m)$ are active, for every $m \in \mathbb{Z}$, and by Lemma 5.3(c) all of them have $f_{\mathrm{S}} = [y = r_d^{(2)}(m')]$. So (Q1) gives $\mathfrak{r} = [y = r_d^{(2)}(m')]$. The row contains no light horizontal wall: that would make y an endpoint of a row frame of some depth e, which is impossible for $e < d$ because y lies in no frame of depth e, for $e = d$ because y is interior to a frame of depth d, and for $e > d$ by Lemma 3.13(a). So (Q2) does not apply on this row, the seeds are consistent, and $\mathfrak{r}$ is as claimed. Every $r_d^{(2)}(m')$ falls under this

second case, and an endpoint row is not of this form, which proves (a) for rows; columns are the same with (Q1), (Q2) for $\mathfrak{c}$.

(b) By Lemma 5.3(d) the roots are the sites $\mathbf{r}_e(m_1, m_2)$. Such a root lies on the row $r_d^{(2)}(m_2)$ only if $e = d$, by Lemma 3.13(d), and $r_d^{(1)}(m_1 + 1) - r_d^{(1)}(m_1) = 4^{d+1}$.

(c) The row $y = r_d^{(2)}(m_2)$ lies in $I_d^{(2)}(m_2)$, so Lemma 3.14 places light vertical walls at (z, y) for the endpoints z of the column frames of depth d. The absence of light horizontal walls was shown above. □

Part (c) is why (Q2) tests the orientation of the wall: the seed $\mathfrak{r} = 0$ must not be placed on root rows, and every root row crosses light vertical walls.

5.4 The law of the background and its rigidity

Proposition 5.7 (The background)**.** *(a) There is a measurable map $\tau \mapsto \mathfrak{b}_\tau$ from typical phases to background configurations, with $\mathfrak{b}_{\tau+a} = T_a \mathfrak{b}_\tau$ for $a \in \mathbb{Z}^2$, such that every legal configuration whose quarters have typical phase τ has background $\mathfrak{b}_\tau$.*

(b) For every typical τ, the configuration with background $\mathfrak{b}_\tau$, arbitrary spins, wire values given by (P1) and (P2) from the spins at the roots, and all switches and free bits equal to 0, is legal.

(c) Every translation-invariant probability measure on X_Φ has the same background marginal $\mathbb{K}$, the image of Haar measure on $\mathbb{Z}_2^2$ under $\tau \mapsto \mathfrak{b}_\tau$. It is concentrated on the backgrounds $\mathfrak{b}_\tau$ with τ typical.

(d) If $\omega \in X_\Phi$ has typical quarter phase and $\omega' \in X_\Phi$ agrees with ω outside a finite set, then ω' has the same background as ω.

Proof. (a) The quarters are Y_τ (Proposition 3.10). The orientation tags are given by Proposition 5.1(a). The event tags are $*$ on edges whose tag is not (L, R) or (B, T) (E1). An edge with tag (L, R) lies between consecutive walls bounding a frame of the owner depth of its row; if the row is a free row of that frame, the event tags are those of Lemma 5.3(c), and otherwise the corridor contains no active site and (E2) forces $(\mathrm{wall}, \mathrm{wall})$ on all its edges. Vertical event tags are the same. The logical colours are $\circ$ off the carriers (K1), (K2), and on the corridors and channels of the grids they are given by Lemma 5.4; a corridor of carriers without active sites copies the colour $\circ$ of the walls at its ends. The background components of the cell records are given by Lemma 5.4, and the markers by Lemma 5.6. Each of these values is a function of the restriction of Y_τ to a finite window whose size depends on τ, so $\tau \mapsto \mathfrak{b}_\tau$ is measurable, and $\mathfrak{b}_{\tau+a} = T_a \mathfrak{b}_\tau$ by uniqueness and Lemma 3.9.

(b) The values described in (a) satisfy (O1)–(Q2) and (C0)–(C7) by construction: each was obtained by solving these rules, and the solutions exist because the corridors and grids are finite. With all switches and free bits 0, (C8), (C9) and (P3) hold. On a marked row, (P1) determines the wire values from the spins at the roots, which by Lemma 5.6(b) recur at finite distance in both directions; on an unmarked row the wires are blank. Columns are the same.

(c) Let μ be a translation-invariant probability measure on X_Φ. Its quarter marginal is a translation-invariant probability measure on the tilings of $\mathbb{A}^{\mathfrak{q}}$, hence by Proposition 3.10(a) the image of Haar measure under $\tau \mapsto Y_\tau$. Haar-almost every τ is typical, and (a) gives the background.

(d) The quarters of ω' form a tiling of $\mathbb{A}^{\mathfrak{q}}$ that agrees with Y_τ outside a finite set, so it equals Y_τ by Proposition 3.10(b). Then (a) applies to ω' with the same τ. □

Part (d) is the frozen-background property: every legal configuration over a typical scaffold, including its payload, admits no finite change of its background. The quarter encoding enters through the global checker phase, which a finite edit cannot change (Proposition 3.10(b)).

Fix a typical τ and write $\mathfrak{b} = \mathfrak{b}_\tau$. Let $P_\mathfrak{b}$ be the set of payload configurations p with $(\mathfrak{b}, p)$ legal. For a finite $\Lambda \subseteq \mathbb{Z}^2$ and $p \in P_\mathfrak{b}$, the *frozen kernel* is

$$\gamma^\mathfrak{b}_\Lambda(p' \mid p) = \frac{\mathbf{1}[p'_{\Lambda^c} = p_{\Lambda^c}]\, e^{-\beta H_\Lambda(\mathfrak{b}, p')}}{\sum_{p'' :\, p''_{\Lambda^c} = p_{\Lambda^c}} e^{-\beta H_\Lambda(\mathfrak{b}, p'')}},$$

and a probability measure ν on payload configurations is *Gibbs for* $\gamma^\mathfrak{b}$ if $\nu(P_\mathfrak{b}) = 1$ and $\nu\gamma^\mathfrak{b}_\Lambda = \nu$ for every finite Λ.

Lemma 5.8 (Conditioning on the background)**.** *Let $X_{\mathrm{typ}} \subseteq X_\Phi$ be the set of legal configurations whose quarters have typical phase.*

(a) For $\omega = (\mathfrak{b}, p) \in X_{\mathrm{typ}}$ and finite Λ, $\gamma_\Lambda(\cdot \mid \omega) = \delta_\mathfrak{b} \otimes \gamma^\mathfrak{b}_\Lambda(\cdot \mid p)$.

(b) For every finite Λ there is a measurable function h_Λ of ω_{Λ^c} such that the background of ω on Λ equals $h_\Lambda(\omega_{\Lambda^c})$ for every $\omega \in X_{\mathrm{typ}}$. So for every probability measure μ on X_{typ}, the σ-algebra generated by the background is contained in the μ-completion of $\mathcal{F}_{\Lambda^c}$.

(c) Let μ be a probability measure on X_{typ}, with background law $\mathbb{L}$ and regular conditional laws $\mu^\mathfrak{b}$ of the payload given the background. Then μ is a Gibbs measure for $\beta\Phi$ if and only if $\mu^\mathfrak{b}$ is Gibbs for $\gamma^\mathfrak{b}$ for $\mathbb{L}$-almost every $\mathfrak{b}$.

Proof. (a) The kernel $\gamma_\Lambda(\cdot \mid \omega)$ charges the legal configurations that agree with ω off Λ. By Proposition 5.7(d) they have background $\mathfrak{b}$, so they are the configurations $(\mathfrak{b}, p')$ with $p' \in P_\mathfrak{b}$ and $p'_{\Lambda^c} = p_{\Lambda^c}$, with weights $e^{-\beta H_\Lambda(\mathfrak{b}, p')}$.

(b) By Lemma 3.9, $\tau \bmod 2^l$ is determined by the quarters in a box B_l, and $\tau(T_a Y) = \tau(Y) + a$, so it is also determined by the quarters in $B_l + a$ for any a. Choosing a with $B_l + a$ disjoint from Λ for each l expresses τ, and hence by Proposition 5.7(a) the background $\mathfrak{b}_\tau$ on Λ, as a measurable function of ω_{Λ^c}.

(c) Suppose μ is Gibbs. Let f be a bounded function of the payload on Λ, h a bounded measurable function of the background, and g a bounded measurable function of the payload off Λ. By (a), the kernel $\gamma_\Lambda(\cdot \mid \omega)$ changes neither the background nor the payload off Λ, so $\gamma_\Lambda(fgh)(\omega) = g(p)h(\mathfrak{b})(\gamma^\mathfrak{b}_\Lambda f)(p)$ for $\omega = (\mathfrak{b}, p)$. The Gibbs identity for $F = fgh$ gives

$$\int \mathbb{L}(d\mathfrak{b})\, h(\mathfrak{b}) \int f\, g\, d\mu^\mathfrak{b} = \int \mathbb{L}(d\mathfrak{b})\, h(\mathfrak{b}) \int (\gamma^\mathfrak{b}_\Lambda f)\, g\, d\mu^\mathfrak{b}.$$

As h varies, the inner integrals agree for $\mathbb{L}$-almost every $\mathfrak{b}$. Taking Λ and f, g in a countable family that determines measures, $\mu^\mathfrak{b}$ is Gibbs for $\gamma^\mathfrak{b}$ for $\mathbb{L}$-almost every $\mathfrak{b}$; also $\mu^\mathfrak{b}(P_\mathfrak{b}) = 1$ almost surely since $\mu(X_\Phi) = 1$. By (b), the same identities say that $\mathbb{E}_\mu[f \mid \mathcal{F}_{\Lambda^c}] = \gamma^\mathfrak{b}_\Lambda f$, so conditioning on the background only uses information outside Λ. Conversely, if $\mu^\mathfrak{b}$ is Gibbs for $\gamma^\mathfrak{b}$ for $\mathbb{L}$-almost every $\mathfrak{b}$, then for bounded measurable F, by (a),

$$\int \gamma_\Lambda F\, d\mu = \int \mathbb{L}(d\mathfrak{b}) \int \gamma^\mathfrak{b}_\Lambda F(\mathfrak{b}, \cdot)\, d\mu^\mathfrak{b} = \int \mathbb{L}(d\mathfrak{b}) \int F(\mathfrak{b}, \cdot)\, d\mu^\mathfrak{b} = \int F\, d\mu. \qquad \square$$

6 The payload is a union of Ising models

Fix a typical τ and write $\mathfrak{b} = \mathfrak{b}_\tau$. This section shows that the Gibbs measures of the frozen kernels $\gamma^\mathfrak{b}$ are in bijection with the Gibbs measures of an Ising model on a disjoint union of square lattices, one for each ready depth.

6.1 Edges and their private coordinates

For $d \geq 1$, the roots of depth d are the sites $\mathbf{r}_d(m_1, m_2)$, $(m_1, m_2) \in \mathbb{Z}^2$ (Lemma 5.3(d)). For each d, m_1, m_2 there are two *edges* with *head* $v = \mathbf{r}_d(m_1, m_2)$: the horizontal edge with *tail* $u = \mathbf{r}_d(m_1 - 1, m_2)$ and the vertical edge with tail $u = \mathbf{r}_d(m_1, m_2 - 1)$. Let $\xi(i, j)$ be the sites of the grid whose root is v, and let $N = N_d$. The *private coordinates* η_e of the horizontal edge e are:

- the *arm*: the wire values $w^{\mathrm{E}}(z, v_2)$ for $u_1 \leq z < v_1$ and $w^{\mathrm{W}}(z, v_2)$ for $u_1 < z \leq v_1$;
- if $\rho_d = 1$, the *sensor*: the components $x_{\mathrm{h}}^-, x_{\mathrm{h}}^+, y_{\mathrm{h}}^-, y_{\mathrm{h}}^+$ of the cell records at $\xi(0, j)$, $j \in [0, N)$, and the components $x_{\mathrm{h}}^-, x_{\mathrm{h}}^+$ of ℓ^{S} and ℓ^{N} at the sites of the column of v from $\xi(0, 0)$ to $\xi(0, N - 1)$.

The private coordinates of the vertical edge are defined in the same way, with the wire values w^{N}, w^{S} along the column of v and the sensor components with subscript v.

Lemma 6.1 (Factorization)**.** *(a) The sets η_e are pairwise disjoint and contain no spin. Every payload coordinate that is neither a spin nor in some η_e takes one fixed value in every $p \in P_{\mathfrak{b}}$.*

(b) For each edge e with tail u and head v of depth d there is a function $W_e(\eta_e, \sigma_u, \sigma_v) \geq 0$ such that, for every finite Λ and $p \in P_{\mathfrak{b}}$, the frozen kernel $\gamma_\Lambda^{\mathfrak{b}}(\cdot \mid p)$ is the kernel of the coordinates at the sites of Λ for the factors W_e, one for each edge, together with the indicators that the coordinates of (a) take their fixed values. Explicitly, with (x^-, x^+) the switches of the sensor of e at $\xi(0, 0)$, $W_e = \mathbf{1}[\mathit{arm} \equiv \sigma_u]$ if $\rho_d = 0$, and if $\rho_d = 1$

$$W_e = \mathbf{1}[\mathit{arm} \equiv \sigma_u]\, \mathbf{1}[\mathit{sensor\ consistent}]\, \mathbf{1}\big[(x^-, x^+) = (0, 1) \Rightarrow \sigma_v = \sigma_u\big]\, e^{-20\beta(x^- + x^+)},$$

where "arm $\equiv \sigma_u$" means that all wire values of the arm equal σ_u, and the sensor is consistent when all copies of $x^\pm$ along the column equal the values at $\xi(0, 0)$ and every free bit $y^\pm(0, j)$ is 0 unless $x^\pm = 1$ and $j < 20n(d) \pm 5$.

Proof. (a) Disjointness: by Lemma 5.6(b), the arms of one root row partition its wire coordinates; different root rows carry different wire coordinates, and horizontal and vertical wire values are different fields. Sensor coordinates of different edges are different components or lie in the first columns of different grids, and the first columns of different grids are disjoint because root columns of different depths are disjoint (Lemma 3.13(d)) and grids of one depth are disjoint.

Fixed coordinates: on an unmarked row all wire values are $\square$ by (P1), and every wire coordinate on a marked row lies in an arm. Outside the first logical column of a grid, (C8) and (C9) set $x = y = 0$ and $\ell_x^{\mathrm{N}} = 0$ at every active site, so by (K2) and matching every x component of a vertical colour there is 0, and at non-carriers the vertical colour is $\circ$. In the first column of a grid with $\rho_d = 0$, (C8) sets $x = 0$ at the root, the copies along the column are then 0, and (C9) gives $y = 0$. These are all the payload coordinates outside the arms, the sensors of ready grids and the spins.

(b) The payload enters only (P1)–(P3), (C8), (C9) and the charge, and these split into conditions of the kinds listed in Section 4.7. We go through them, given the background.

- (P1) on a root row: at the tail root $w^{\mathrm{E}}(u) = \sigma_u$, and at every other site $w^{\mathrm{E}} = w^{\mathrm{W}}$; with matching across edges, the arm of e has all values equal to σ_u, and all values lie in $\{+1, -1\}$. These are conditions on the arm and σ_u.
- (C8) along the first column of a ready grid, with (K2) along its channels: the copies of each switch equal its value at the root, and the root value is free. These are conditions on one switch of one sensor.

- (C9): $y_k^{\pm}(0,j)$ is free if $x_k^{\pm}=1$ and $g^{\pm}(0,j)=1$, and 0 otherwise; by Lemma 5.4(d) and Lemma 5.5(a), $g^{\pm}(0,j)=[j<20n(d)\pm 5]$.
- (P3) at v: if $(x^-,x^+)=(0,1)$ then $\sigma_v=w^{\mathrm{W}}(v)$, and $w^{\mathrm{W}}(v)$ is the last value of the arm, which equals σ_u.
- The charge: by Lemma 5.4(a), $\mathrm{rc}(0,j)=\min(j,80)$, and $N_d\geq 80$ when $\rho_d=1$, since then $d\geq 8$. So the switch x_{h}^- is charged at the 20 cells $j\in[0,20)$, x_{h}^+ at $j\in[20,40)$, x_{v}^- at $j\in[40,60)$ and x_{v}^+ at $j\in[60,80)$, each copy contributing the value of the switch.

The frozen Boltzmann weight is therefore the product of the factors W_e and of the indicators of (a), up to factors that do not involve the coordinates at the sites of Λ; these cancel in the kernel. Spins at sites that are not roots, and spins at roots apart from the factors above, appear in no condition. □

6.2 The sensor sum

Lemma 6.2 (The sensor sum)**.** *Let e be an edge with head of depth d, tail spin σ_u and head spin σ_v, and let $Z_e(\sigma_u,\sigma_v)=\sum_{\eta_e}W_e(\eta_e,\sigma_u,\sigma_v)$. If $\rho_d=0$, then $Z_e=1$. If $\rho_d=1$, put $n=n(d)$ and $s=\beta/\log 2$. Then*

$$Z_e(\sigma_u,\sigma_v)=H_n+G_n\,\mathbf{1}[\sigma_u=\sigma_v],\qquad G_n=2^{20n+5}e^{-20\beta},\quad H_n=1+2^{20n-5}e^{-20\beta}+2^{40n}e^{-40\beta},$$

so that $Z_e(\sigma_u,\sigma_v)=C_n\exp\bigl(J_n(\beta)\sigma_u\sigma_v\bigr)$ for a constant $C_n>0$, with

$$e^{2J_n(\beta)}=1+\frac{1}{b_n(\beta)},\qquad b_n(\beta)=\frac{H_n}{G_n}=2^{20(n-\frac14-s)}+2^{20(s-n-\frac14)}+2^{-10},$$

that is, $b_n(\beta)=2^{-4}\cosh\bigl(20(\beta-n\log 2)\bigr)+2^{-10}$.

Proof. If $\rho_d=0$, the private coordinates are the arm, which has exactly one admissible value. Let $\rho_d=1$. The arm again has one admissible value. Given the switches (x^-,x^+), the consistent sensor configurations are determined by the free bits: $y^-(0,j)$ is free for the $20n-5$ values $j<20n-5$ if $x^-=1$, and $y^+(0,j)$ is free for the $20n+5$ values $j<20n+5$ if $x^+=1$; all these values of j are below N_d by Lemma 5.5(a). So the switch values (x^-,x^+) contribute

$$(0,0)\colon 1,\qquad (1,0)\colon 2^{20n-5}e^{-20\beta},\qquad (0,1)\colon 2^{20n+5}e^{-20\beta}\,\mathbf{1}[\sigma_u=\sigma_v],\qquad (1,1)\colon 2^{40n}e^{-40\beta}.$$

Their sum is $H_n+G_n\mathbf{1}[\sigma_u=\sigma_v]$. Hence $Z_e(+,+)/Z_e(+,-)=1+G_n/H_n=e^{2J_n}$, and $Z_e=C_ne^{J_n\sigma_u\sigma_v}$ with $C_n=\sqrt{H_n(H_n+G_n)}$. With $e^{-20\beta}=2^{-20s}$,

$$\frac{H_n}{G_n}=2^{-20n-5+20s}+2^{-10}+2^{20n-5-20s},$$

which is the first formula for b_n. Writing $2^{20(n-s)}+2^{20(s-n)}=2\cosh(20(n-s)\log 2)$ and $20(n-s)\log 2=20(n\log 2-\beta)$ gives the second. □

The numbers are chosen so that the sensor resonates: $2^{20n\pm5}$ assignments of the $20n\pm5$ free bits against the energy 20 of one switch. The coupling $J_n(\beta)$ is large only when β is close to $n\log 2$.

6.3 Eliminating the private coordinates

Lemma 6.3 (Elimination). *Let V be a countable set of spins with values in $\{-1,+1\}$ and E a set of edges e, each with a tail $u(e)$ and a head $v(e)$ in V, such that every spin lies in finitely many edges. For each e let η_e be a finite set of coordinates with finite value sets, the sets η_e being pairwise disjoint and disjoint from V, and let $W_e(\eta_e, \sigma_{u(e)}, \sigma_{v(e)}) \geq 0$ with $Z_e(\sigma_u, \sigma_v) = \sum_{\eta_e} W_e > 0$ for all spin values. Let γ be the specification of the factors W_e on the coordinates $V \cup \bigcup_e \eta_e$, and $\bar{\gamma}$ that of the factors Z_e on V. For a measure ν on spin configurations put*

$$L(\nu)(d\sigma, d\eta) = \nu(d\sigma) \prod_{e \in E} K_e(\eta_e \mid \sigma_{u(e)}, \sigma_{v(e)}), \qquad K_e = \frac{W_e}{Z_e}.$$

Then L is a bijection from the Gibbs measures of $\bar{\gamma}$ onto the Gibbs measures of γ, and its inverse is the marginal on the spins.

Proof. For a finite set $T \subseteq V$ let $E(T)$ be the set of edges with an endpoint in T, and for a finite set $E' \subseteq E$ let $S(T, E')$ consist of T and the private coordinates of the edges in $E(T) \cup E'$. Every finite set of coordinates lies in such a set: take for T its spins and for E' the edges whose private coordinates it meets. By Lemma 2.3 it suffices to check the Gibbs identities for the sets $S = S(T, E')$.

Let ν be Gibbs for $\bar{\gamma}$ and $\mu = L(\nu)$. Under μ, given all spins, the η_e are independent with laws K_e. Condition on all coordinates outside S: these are the spins outside T and the private coordinates of edges not in $E(T) \cup E'$, which depend only on spins outside T. So the conditional law of the coordinates in S is

$$\bar{\gamma}_T(\sigma_T \mid \sigma_{T^c}) \prod_{e \in E(T) \cup E'} K_e(\eta_e \mid \sigma) \;\propto\; \prod_{e \in E(T)} Z_e \prod_{e \in E(T)} \frac{W_e}{Z_e} \prod_{e \in E' \setminus E(T)} \frac{W_e}{Z_e} \;\propto\; \prod_{e \in E(T) \cup E'} W_e,$$

because Z_e is constant for $e \notin E(T)$. This is γ_S, so μ is Gibbs for γ.

Conversely let μ be Gibbs for γ. Applying the identity for $S(T, \emptyset)$ and summing over the private coordinates of $E(T)$ shows that the conditional law of σ_T given all coordinates outside $S(T, \emptyset)$ is $\prod_{e \in E(T)} Z_e$, normalized, which depends only on σ_{T^c}; so the spin marginal ν is Gibbs for $\bar{\gamma}$. Applying the identity for $S(\emptyset, E')$ shows that, given all spins and the private coordinates of the other edges, the private coordinates of the edges in E' are independent with laws K_e. Since E' is an arbitrary finite set, $\mu = L(\nu)$. □

Corollary 6.4 (The effective model). *Let $\Gamma_{\mathfrak{b}}$ be the graph with vertex set $\mathbb{Z}^2$ whose edges are the pairs $\{\mathbf{r}_d(m_1, m_2), \mathbf{r}_d(m_1 + 1, m_2)\}$ and $\{\mathbf{r}_d(m_1, m_2), \mathbf{r}_d(m_1, m_2 + 1)\}$ for the ready depths d, with coupling $J_{n(d)}(\beta)$ on the edges of depth d. The Gibbs measures for $\gamma^{\mathfrak{b}}$ are exactly the measures $L(\nu)$, completed by the fixed values of Lemma 6.1(a), where ν ranges over the Gibbs measures of the Ising model on $\Gamma_{\mathfrak{b}}$. The graph $\Gamma_{\mathfrak{b}}$ is a disjoint union of one square lattice for each ready depth and of isolated vertices.*

Proof. By Lemma 2.3 and Lemma 6.1(b), Gibbs measures for $\gamma^{\mathfrak{b}}$ are the Gibbs measures for the factors W_e on the coordinates of all sites, with the fixed coordinates at their values. A root is the head of two edges and the tail of two, and other spins lie in no edge, so Lemma 6.3 applies. By Lemma 6.2, the factor Z_e is constant for edges of non-ready depths and proportional to $e^{J_{n(d)}\sigma_u\sigma_v}$ for ready ones; constant factors do not change a specification. Roots of different depths are different sites, and the roots of depth d with these edges form a square lattice. □

The square lattice of depth d is $\{\mathbf{r}_d(m) : m \in \mathbb{Z}^2\}$ with its nearest-neighbour edges, and $m \mapsto \mathbf{r}_d(m)$ is a graph isomorphism from $\mathbb{Z}^2$. A translation of the background by a maps it onto the lattice of depth d of $T_a\mathfrak{b}$, and the two isomorphisms differ by a translation of $\mathbb{Z}^2$.

7 Proof of Theorem A and Corollary B

7.1 Temperatures

Lemma 7.1 (Rational windows)**.** *Let* $\lambda_K = 2\sum_{j=0}^{K-1} \frac{1}{(2j+1)3^{2j+1}}$, $\beta_n = n\lambda_{n+3}$ *and* $J_n = (\beta_n - \frac{1}{1000}, \beta_n + \frac{1}{1000})$.

(a) $0 < \log 2 - \lambda_K \le \frac{3}{4(2K+1)9^K}$ *for* $K \ge 1$.

(b) $0 < n\log 2 - \beta_n \le \frac{3}{26244} < \frac{1}{1000}$ *for* $n \ge 1$.

(c) $\beta_{n+1} - \beta_n > \frac{2}{3}$, *and* $\beta_1 - \frac{1}{1000} > 0$; *so the intervals* J_n *are pairwise disjoint and consist of positive numbers.*

(d) If $\beta \in J_n$, *then* $|\beta/\log 2 - n| < \frac{1}{250}$.

Proof. (a) $\log 2 = \log\frac{1+1/3}{1-1/3} = 2\sum_{j\ge 0}\frac{1}{(2j+1)3^{2j+1}}$, a series of positive terms. Its tail after K terms is at most $\frac{2}{(2K+1)3^{2K+1}}\sum_{i\ge 0} 9^{-i} = \frac{2\cdot 9}{8(2K+1)\cdot 3\cdot 9^K} = \frac{3}{4(2K+1)9^K}$.

(b) By (a), $0 < n(\log 2 - \lambda_{n+3}) \le \frac{3}{4}\, n\, 9^{-n-3}$. The sequence $n9^{-n}$ decreases for $n \ge 1$, so the bound is at most $\frac{3}{4}\cdot 9^{-4} = \frac{3}{26244}$, and $26244 > 3000$.

(c) $\beta_{n+1} - \beta_n = \lambda_{n+4} + n(\lambda_{n+4} - \lambda_{n+3}) > \lambda_{n+4} \ge \lambda_1 = \frac{2}{3}$, and $\beta_1 = \lambda_4 > \frac{2}{3}$.

(d) By (b), $|\beta - n\log 2| < \frac{1}{1000} + \frac{3}{26244} < \frac{2}{1000}$, and $\log 2 > \frac{1}{2}$. □

Lemma 7.2 (Thermal bounds)**.** *Let* $\beta > 0$, $s = \beta/\log 2$ *and* $m \ge 1$, *and let* b_m, J_m *be as in Lemma 6.2.*

(a) If $|s - m| < \frac{1}{20}$, *then* $b_m(\beta) \le \frac{81}{1024} < \frac{1}{8}$ *and* $e^{J_m(\beta)} > 3$.

(b) If $|s - m| \ge \frac{19}{20}$, *then* $b_m(\beta) \ge 2^{14}$ *and* $4\tanh J_m(\beta) \le \frac{4}{32769} < 1$.

(c) If $|s - m| > \frac{9}{20}$, *then* $b_m(\beta) > 16$ *and* $4\tanh J_m(\beta) < \frac{4}{33} < 1$.

Proof. Write $b_m = 2^{-5}(u + u^{-1}) + 2^{-10}$ with $u = 2^{20(s-m)}$. (a) Here $\frac{1}{2} < u < 2$, so $u + u^{-1} < \frac{5}{2}$ and $b_m < \frac{80}{1024} + \frac{1}{1024}$. Then $e^{2J_m} = 1 + 1/b_m > 9$. (b) One of u, u^{-1} equals $2^{20|s-m|} \ge 2^{19}$, so $b_m \ge 2^{14}$. From $e^{2J} = 1 + 1/b$ we get $\tanh J = \frac{1}{2b+1} \le \frac{1}{2^{15}+1}$. (c) As in (b), $b_m > 2^{9-5} = 16$ and $\tanh J_m < \frac{1}{33}$. □

7.2 The main theorem

Let $\mathcal{D}_n = \{d \ge 1 : \rho_d = 1,\ n(d) = n\}$ be the set of ready depths with label n. It does not depend on the background. By Lemma 5.5, $\mathcal{D}_n$ is empty if U does not halt on n and infinite if it does.

Theorem 7.3. *Let* Φ *be the interaction of Definition 4.1,* $n \ge 1$ *and* $\beta \in J_n$.

(i) If U *does not halt on* n, *then* $\beta\Phi$ *has exactly one translation-invariant Gibbs measure.*

(ii) If U *halts on* n, *then* $\beta\Phi$ *has at least two translation-invariant Gibbs measures.*

Moreover, for every $\beta > 0$, $\beta\Phi$ *has uncountably many Gibbs measures that are not translation invariant.*

Proof. Throughout, τ ranges over typical phases, $\mathfrak{b}_\tau$ is the background of Proposition 5.7, $\Gamma_\tau = \Gamma_{\mathfrak{b}_\tau}$ is the graph of Corollary 6.4, and L is the lift of Lemma 6.3, completed by the fixed coordinates. Let $s = \beta/\log 2$; by Lemma 7.1(d), $|s - n| < \frac{1}{250}$, so $|s - m| > \frac{19}{20}$ for every integer $m \ne n$.

Step 1: every translation-invariant Gibbs measure is an average over backgrounds. Let μ be a translation-invariant Gibbs measure. By Proposition 5.7(c) its background marginal is $\mathbb{K}$, so μ is concentrated on X_{typ}, and by Lemma 5.8(c) and Corollary 6.4,

$$\mu = \int \delta_{\mathfrak{b}_\tau} \otimes L(\nu^\tau)\, \mathrm{Haar}(d\tau) \tag{2}$$

for a measurable family $\tau \mapsto \nu^\tau$ in which ν^τ is a Gibbs measure of the Ising model on Γ_τ for almost every τ. Conversely, by Lemma 5.8(c), every such integral is a Gibbs measure for $\beta\Phi$, and it is translation invariant if $\nu^{\tau+a} = T_a\nu^\tau$ for all $a \in \mathbb{Z}^2$ and almost every τ, because $\mathfrak{b}_{\tau+a} = T_a\mathfrak{b}_\tau$ and Haar measure is invariant.

Step 2: case (i). If U does not halt on n, every ready depth d has $n(d) \neq n$, so every edge of Γ_τ has a coupling $J_{n(d)}(\beta)$ with $4\tanh J_{n(d)}(\beta) \leq \frac{4}{32769}$ by Lemma 7.2(b). Every vertex of Γ_τ has at most four neighbours. By Lemma 2.5, the Ising model on Γ_τ has exactly one Gibbs measure ν^τ. Its finite-dimensional marginals are limits of finite-volume kernels with any fixed boundary condition, so they are measurable in τ; and $\nu^{\tau+a} = T_a\nu^\tau$, since $T_a\nu^\tau$ is a Gibbs measure for $\Gamma_{\tau+a}$, which is the translate of Γ_τ. By Step 1, every translation-invariant Gibbs measure equals (2) with this family, and that integral is a translation-invariant Gibbs measure. So there is exactly one.

Step 3: case (ii). Let U halt on n, so $\mathcal{D}_n$ is infinite. The graph Γ_τ is the disjoint union of the square lattices Γ_τ^d of depths $d \in \mathcal{D}_n$ and of a graph $\Gamma_\tau^{\mathrm{rest}}$, formed by the lattices of the other ready depths and the isolated vertices. On Γ_τ^d, $d \in \mathcal{D}_n$, the coupling is $J_n(\beta)$ with $e^{J_n(\beta)} > 3$ by Lemma 7.2(a). Let $\nu^{\tau,d,\pm}$ be the image of the plus and minus states of Lemma 2.4 under the isomorphism $m \mapsto \mathbf{r}_d(m)$; it does not depend on the choice of the isomorphism, because the plus and minus states are translation invariant. On $\Gamma_\tau^{\mathrm{rest}}$ every coupling satisfies $4\tanh J \leq \frac{4}{32769}$, so it has exactly one Gibbs measure $\nu^{\tau,\mathrm{rest}}$ by Lemma 2.5. Put

$$\nu^{\tau,\pm} = \nu^{\tau,\mathrm{rest}} \otimes \bigotimes_{d\in\mathcal{D}_n} \nu^{\tau,d,\pm}.$$

A product of Gibbs measures on the components of a graph is a Gibbs measure on the graph, because the kernel of a finite set is the product of the kernels of its intersections with the components. The family is measurable in τ and satisfies $\nu^{\tau+a,\pm} = T_a\nu^{\tau,\pm}$. By Step 1, $\mu^\pm = \int \delta_{\mathfrak{b}_\tau} \otimes L(\nu^{\tau,\pm})\, \mathrm{Haar}(d\tau)$ are translation-invariant Gibbs measures.

They are different. Fix $d \in \mathcal{D}_n$ and let E be the event that the origin is a root of depth d. It is the event $\tau_i \equiv -4^d - 2 \pmod{4^{d+1}}$ for $i = 1, 2$ (Lemma 3.13(d)), so it has Haar probability $4^{-2(d+1)} > 0$, and by Lemma 3.9 it is determined by the quarters in a finite box. On E, Lemma 2.4 gives $\nu^{\tau,+}(\sigma_0 = +1) \geq \frac{2}{3}$ and $\nu^{\tau,-}(\sigma_0 = +1) \leq \frac{1}{3}$. Hence

$$\mu^+\big(E \cap \{\sigma_0 = +1\}\big) - \mu^-\big(E \cap \{\sigma_0 = +1\}\big) \geq \tfrac{1}{3}\, 4^{-2(d+1)} > 0,$$

and the two measures differ on an event determined by finitely many sites.

Step 4: Gibbs measures that are not translation invariant. Let $\beta > 0$ and τ typical. The Ising model on Γ_τ has a Gibbs measure ν, by compactness. By Lemma 5.8(c) with the background law $\delta_{\mathfrak{b}_\tau}$, the measure $\delta_{\mathfrak{b}_\tau} \otimes L(\nu)$ is a Gibbs measure for $\beta\Phi$. Its translate by $a \neq 0$ has background $\mathfrak{b}_{\tau+a}$, whose quarters $Y_{\tau+a}$ differ from Y_τ because their phases differ. So it is not translation invariant, and different typical τ give different measures. □

Theorem A is Theorem 7.3 together with Lemma 7.1: the interaction has a finite site set, is translation invariant and nearest-neighbour, takes values in $\{0, 1, +\infty\}$, and contains U only through the table of M.

Corollary 7.4. *The set $\mathcal{U} = \{\beta \in \mathbb{Q}_{>0} : \beta\Phi$ has exactly one translation-invariant Gibbs measure$\}$ is not decidable. The map $n \mapsto \beta_n$ is computable, and $\beta_n \in \mathcal{U}$ if and only if U does not halt on n.*

Proof. The numbers β_n are rational and computable from n by their defining finite sum, and $\beta_n \in J_n$. By Theorem 7.3, $\beta_n \in \mathcal{U}$ exactly when $n \notin \mathcal{H}_U$. An algorithm deciding $\mathcal{U}$ would decide $\mathcal{H}_U$. □

7.3 Further properties

Remark 7.5 (Windows of uniqueness). If $|\beta/\log 2 - (n + \frac{1}{2})| < \frac{1}{20}$ for an integer $n \geq 0$, then every integer label m has $|\beta/\log 2 - m| > \frac{9}{20}$, so by Lemma 7.2(c) and the argument of Step 2, $\beta\Phi$ has exactly one translation-invariant Gibbs measure, whatever U does. The halting set $\mathcal{H}_U$ is infinite, since it is undecidable. So the set of β with at least two translation-invariant Gibbs measures, which meets J_n for every $n \in \mathcal{H}_U$, and the set with exactly one, which contains these windows, both have infinitely many connected components.

Remark 7.6 (Spin-flip symmetry). The map that changes the sign of every spin and of every non-blank wire value, and fixes all other fields, preserves every rule and the charge: equality and copying of wire values, the reset $w^{\mathrm{E}} = \sigma$, $w^{\mathrm{N}} = \sigma$ at roots, blank wire values, and the alignment conditions of (P3) are all invariant. This holds at every site value, including site values that violate other rules. So the interaction is invariant under the global spin flip, term by term, and so is any interaction obtained from it by replacing $+\infty$ with a finite constant. The proof of Theorem 7.3 does not apply to such a replacement.

8 The exact phase diagram, assuming the Ising solution

Theorem A uses only the Peierls argument and Dobrushin's theorem. With the exact solution of the two-dimensional Ising model as an additional input, the whole set of inverse temperatures with coexisting phases can be described. This section is conditional on the following classical facts.

Input (I). For the Ising model on $\mathbb{Z}^2$ with coupling $J \geq 0$, put $J_c = \frac{1}{2}\log(1 + \sqrt{2})$. If $J \leq J_c$, the model has exactly one Gibbs measure. If $J > J_c$, the plus state of Lemma 2.4 satisfies $\nu^+(\sigma_0) > 0$.

Kramers and Wannier identified the self-dual value J_c as the candidate critical point [19], and Onsager's exact solution established it [27]; a modern proof that the self-dual point is critical is [4]. The spontaneous magnetization $\nu^+(\sigma_0)$ was computed by Yang [33]; it is positive for $J > J_c$ and zero for $J \leq J_c$. Uniqueness of the Gibbs measure when $\nu^+(\sigma_0) = 0$, including at $J = J_c$, follows from the FKG inequality [13, Chapters 3 and 6], and every Gibbs measure is a mixture of ν^+ and ν^- by the theorem of Aizenman and Higuchi [1, 17]; the full statement is recalled, for instance, in [8, Section 1]. We use only Input (I) as stated.

Lemma 8.1 (Product of unique components)**.** *Let G be a countable graph with bounded degrees that is a disjoint union of subgraphs G_k, and suppose the Ising model on each G_k, with given couplings, has exactly one Gibbs measure ν_k. Then the Ising model on G has exactly one Gibbs measure, $\bigotimes_k \nu_k$.*

Proof. The product is a Gibbs measure, as in Step 3 of the proof of Theorem 7.3. Let μ be a Gibbs measure on G and fix k. For a finite set T of vertices of G_k, the kernel γ_T depends only on the spins of G_k outside T. So the regular conditional law of the spins of G_k given those of $G \setminus G_k$ satisfies the Gibbs identities of G_k for almost every condition, for a countable family

of sets T and test functions, and therefore equals ν_k. Hence the spins of G_k are independent of the rest with law ν_k. Applying this to finitely many components in turn shows that every finite-dimensional marginal of μ is that of the product. □

Proposition 8.2 (The exact diagram)**.** *Assume Input (I) and put* $\Delta = \frac{1}{20}\operatorname{arcosh}\bigl(8\sqrt{2} - \frac{1}{64}\bigr)$*. Then* $0.15 < \Delta < \frac{1}{4}\log 2$*, and for* $\beta > 0$ *the interaction* $\beta\Phi$ *has at least two translation-invariant Gibbs measures if* $\beta \in \mathcal{N}$ *and exactly one otherwise, where*

$$\mathcal{N} = \bigcup_{n \in \mathcal{H}_U} \bigl(n\log 2 - \Delta,\ n\log 2 + \Delta\bigr).$$

The set $\mathcal{N} \cap \mathbb{Q}_{>0}$ *is* Σ_1^0*-complete, and its complement in* $\mathbb{Q}_{>0}$ *is* Π_1^0*-complete.*

Proof. By Lemma 6.2, $J_m(\beta) > J_c$ if and only if $e^{2J_m} > 1 + \sqrt{2}$, that is $b_m(\beta) < 2^{-1/2}$, that is

$$\cosh\bigl(20(\beta - m\log 2)\bigr) < 2^4\bigl(2^{-1/2} - 2^{-10}\bigr) = 8\sqrt{2} - \tfrac{1}{64},$$

that is $|\beta - m\log 2| < \Delta$. Since $8\sqrt{2} - \frac{1}{64} > 11$ and $\operatorname{arcosh} 11 = \log(11 + \sqrt{120}) > \log 21 > 3$, we have $\Delta > \frac{3}{20}$. Since $8\sqrt{2} - \frac{1}{64} < 16 + \frac{1}{64} = \cosh(5\log 2)$, we have $\Delta < \frac{1}{4}\log 2$. So for each β at most one label m has $J_m(\beta) > J_c$, and $J_m(\beta) = J_c$ exactly at $\beta = m\log 2 \pm \Delta$.

Let $\beta \in (n\log 2 - \Delta, n\log 2 + \Delta)$ with $n \in \mathcal{H}_U$. The set $\mathcal{D}_n$ of ready depths with label n is nonempty, and the lattices of these depths have coupling $J_n(\beta) > J_c$. All other components of Γ_τ have couplings at most J_c, or are isolated vertices, and have exactly one Gibbs measure by Input (I); by Lemma 8.1 so does their union. The construction of Step 3 of the proof of Theorem 7.3 gives two translation-invariant Gibbs measures, which differ on the event $E \cap \{\sigma_0 = +1\}$ because $\nu^+(\sigma_0 = +1) = \frac{1}{2}(1 + \nu^+(\sigma_0)) > \frac{1}{2}$ and $\nu^-(\sigma_0 = +1) < \frac{1}{2}$.

Let $\beta \notin \mathcal{N}$. A ready depth d has $n(d) \in \mathcal{H}_U$ by Lemma 5.5(a), so $J_{n(d)}(\beta) \le J_c$. Every component of Γ_τ has exactly one Gibbs measure by Input (I), and so does Γ_τ by Lemma 8.1. Step 2 of the proof of Theorem 7.3, with Lemma 8.1 in place of Lemma 2.5, gives exactly one translation-invariant Gibbs measure.

Degree. A rational $\beta > 0$ lies in $\mathcal{N}$ if and only if there are n and t such that U halts on n within t steps and $|\beta - n\log 2| < \Delta$. The last condition is a strict inequality between computable reals, which is semidecidable; so $\mathcal{N} \cap \mathbb{Q}_{>0}$ is computably enumerable. By Lemma 7.1(b), $|\beta_n - n\log 2| < \frac{1}{1000} < \Delta$, and β_n is at distance more than $\log 2 - \frac{1}{1000} > \Delta$ from $m\log 2$ for every $m \neq n$; so $\beta_n \in \mathcal{N}$ if and only if $n \in \mathcal{H}_U$. Since $\mathcal{H}_U$ is Σ_1^0-complete, so is $\mathcal{N} \cap \mathbb{Q}_{>0}$, and its complement in $\mathbb{Q}_{>0}$ is Π_1^0-complete. □

At the endpoints $n\log 2 \pm \Delta$ the coupling equals J_c, and uniqueness there uses the critical case of Input (I). The degree statement uses only open intervals and does not need to decide whether a rational number is an endpoint.

9 Open problems

Finite energies. The infinite energies freeze the background and keep the computations exact. With finite energies every local pattern has positive probability, the background can fluctuate, and errors can imitate a ready grid or break a wire.

Open problem 9.1. Is there one translation-invariant finite-range interaction on $\mathbb{Z}^d$ with finite rational energies, and computable rational inverse temperatures β_n, such that uniqueness of the translation-invariant Gibbs measure at β_n is undecidable?

All Gibbs measures. The model of Theorem A has Gibbs measures that are not translation invariant at every temperature, so its undecidable property is uniqueness among translation-invariant Gibbs measures.

Open problem 9.2. Is there one interaction, with or without hard constraints, for which uniqueness among all Gibbs measures, as a function of β, is undecidable?

Smaller models. Our alphabet is large, the model is two-dimensional, and its rules are built to order. One dimension is excluded, hard constraints included. Recode a finite-range interaction on $\mathbb{Z}$ with a finite alphabet as a transfer matrix $W(\beta)$ on a finite graph, with entries $e^{-\beta E}$ on the allowed edges and 0 on the forbidden ones; the allowed edges do not depend on β. A bi-infinite legal path crosses between strongly connected components of this graph at most a bounded number of times, so a translation-invariant measure gives a crossing at the origin probability 0, and a translation-invariant Gibbs measure is a mixture of translation-invariant Gibbs measures carried by single components. On a component that contains a cycle the Perron–Frobenius theorem gives exactly one, the uniform average over the cyclic classes when the component is periodic. So the number of extremal translation-invariant Gibbs measures equals the number of components that contain a cycle, at every finite $\beta > 0$, and uniqueness cannot change with the temperature.

Open problem 9.3. Is there a two-dimensional example with a small alphabet or with a natural interaction?

How this work was done

The authors chose the question, directed the work and judged every result. AI models from two developers derived, checked and wrote the mathematics. The theorem and its proof were derived by OpenAI's GPT-6.1 Sol. Three agents of the same developer each checked the complete written proof, in separate runs that did not write it. Anthropic's Claude Opus 5.5 then re-derived the proof refutation-first, in two rounds: an independent derivation by hand, followed by an adversarial attack on its verdicts by a separate run that did not write them. This review found no false step and recommended three changes of presentation, which this manuscript makes: the passage from the quartered encoding to finite rigidity (Proposition 3.10(b)), the lemma that justifies conditioning on the background (Lemma 5.8), and the orientation of the seed of the row markers (rule (Q2) and Lemma 5.6(c)). The search of the prior literature was done by Claude Opus 5.5. The manuscript was written by Claude Opus 5.5 from the audited proofs; this includes the explicit rules of Section 4 and the program of Section C, which the proofs specified in words. A separate run of OpenAI's GPT-6.1 Sol, which had not seen the work, then reviewed the finished manuscript refutation-first, including the explicit rules, the tables and the program, and compared it with the original derivation. It checked finite parts with pairs of independently written programs: the scaffold tables and their seams, the program of M on bounded inputs, and the arithmetic of the windows. It found no false step in the main results, and five local errors and five points of presentation or credit, which this manuscript corrects. Models are not listed as authors; the authors take responsibility for every claim.

Verification status

- **Proved in the text.** Theorem A (Theorem 7.3 with Lemma 7.1), Corollary B (Corollary 7.4) and every lemma and proposition of Sections 3 and 5 to 7 and of Sections A and C, given the inputs below.
- **Imported with stated scope.** Existence and translation invariance of the plus and minus states of the Ising model on $\mathbb{Z}^2$, through the FKG inequality [13]; Dobrushin's uniqueness theorem [11, 15]; and standard measure theory (regular conditional probabilities, compactness of the configuration space).
- **Conditional.** Proposition 8.2 assumes Input (I) of Section 8, the classical exact facts about the two-dimensional Ising model. Theorem A and Corollary B do not use it.
- **The finite tables.** The interaction is given by explicit rules on a finite product of finite sets (Section 4, Section B); the machine M is given by its program (Section C), and its transition table contains the transition table of U. The tables were not listed in full. The rules were written for this manuscript from the specification of the audited proofs. Their agreement with that specification was checked by the writer; then by hand and refutation-first in a separate two-round review of this manuscript by Claude Opus 5.5, which derived every new rule and the program again and then attacked its own verdicts; and then in the final review by GPT-6.1 Sol. No program listed or checked the whole allowed set, which depends on U.
- **Checked by machine.** No proof uses a computation. In the final review, independently written programs checked finite parts of the construction: the scaffold tables and their seams, the program of M on bounded inputs, and the arithmetic of the windows. These checks support the text; they are not part of any proof.
- **Not formalized.** Nothing is verified in Lean.
- **Internal only.** Every review is internal; none of it is journal peer review.

A The scaffold: tables and proofs

This appendix lists the finite data of Section 3, Ollinger's substitution [26] in the transcription of [21] with the ruler decoration of [12], and proves Proposition 3.3, Lemma 3.13(a)–(c) and Lemma 3.14. The arguments follow [12]; they compare components of tiles, so no list of cases is needed. Vertical atoms are $\mathsf{r}_0, \mathsf{r}_2, \mathsf{g}_0, \mathsf{g}_2$ and horizontal atoms $\mathsf{r}_1, \mathsf{r}_3, \mathsf{g}_1, \mathsf{g}_3$.

A.1 Tables

The left table shows: each corner has exactly the four endpoints of one thin type, namely $F(\mathsf{a}) = \angle\mathsf{b}$, $F(\mathsf{b}) = \angle\mathsf{d}$, $F(\mathsf{c}) = \angle\mathsf{c}$, $F(\mathsf{d}) = \angle\mathsf{a}$; thin types match horizontally exactly in the pairs (a,c), (c,a), (b,d), (d,b), west tile first, and vertically exactly in the pairs (a,d), (d,a), (b,c), (c,b), south tile first. The black maps of the substitution are

ψ	A	B	C	D	E
$\mathrm{se}(\psi)$	B	B	C	C	C
$\mathrm{nw}(\psi)$	E	D	D	D	E

	∂_N	∂_S	∂_E	∂_W
thin a	r_0	g_2	r_1	g_3
thin b	g_0	r_2	g_1	r_3
thin c	r_2	g_0	g_3	r_1
thin d	g_2	r_0	r_3	g_1
corner $\angle$a	g_2	r_0	r_3	g_1
corner $\angle$b	r_0	g_2	r_1	g_3
corner $\angle$c	r_2	g_0	g_3	r_1
corner $\angle$d	g_0	r_2	g_1	r_3
line $\langle\alpha,\alpha'\rangle$	α	α	α'	α'

	∂_N	∂_S	∂_E	∂_W
black A	0	1	0	1
black B	1	0	0	0
black C	1	0	1	1
black D	1	1	1	0
black E	0	0	1	0

Table 2: Endpoints of the thin and thick components, and bits of the black component. A line $\langle\alpha,\alpha'\rangle$ has a vertical atom α and a horizontal atom α'; all sixteen lines occur.

They satisfy $\partial_W \operatorname{se}(\psi) = \partial_E\psi$, $\partial_S \operatorname{nw}(\psi) = \partial_N\psi$, $\partial_E \operatorname{se}(\psi) = \partial_E\psi$, $\partial_N \operatorname{nw}(\psi) = \partial_N\psi$, $\partial_N \operatorname{se}(\psi) = 1$ and $\partial_E \operatorname{nw}(\psi) = 1$ for every ψ; and $\operatorname{se}(\psi)$ is the type in $\{\mathsf{B},\mathsf{C}\}$ with west bit $\partial_E\psi$, $\operatorname{nw}(\psi)$ the type in $\{\mathsf{D},\mathsf{E}\}$ with south bit $\partial_N\psi$.

thin	thick	black	condition	tiles
a	corner	A	none	4
a	line $\langle\alpha,\alpha'\rangle$	B	$\alpha \neq r_0$	12
a	line $\langle\alpha,\alpha'\rangle$	C	$\alpha \neq r_2$	12
a	line $\langle\alpha,\alpha'\rangle$	D	$\alpha' \neq r_3$	12
a	line $\langle\alpha,\alpha'\rangle$	E	$\alpha' \neq r_1$	12
b	corner	A	none	4
c	line $\langle\alpha,\alpha'\rangle$	B	$\alpha \neq r_0$	12
c	line $\langle\alpha,\alpha'\rangle$	C	$\alpha \neq r_2$	12
d	line $\langle\alpha,\alpha'\rangle$	D	$\alpha' \neq r_3$	12
d	line $\langle\alpha,\alpha'\rangle$	E	$\alpha' \neq r_1$	12
			total	104

Table 3: The 104 tiles of $\mathbb{O}$.

child	thin	thick	black
$(0,0)$	a	κ	ψ
$(1,0)$	c	$\langle\partial_S\theta,\ \partial_E\kappa\rangle$	$\operatorname{se}(\psi)$
$(0,1)$	d	$\langle\partial_N\kappa,\ \partial_W\theta\rangle$	$\operatorname{nw}(\psi)$
$(1,1)$	b	$F(\theta)$	A

child	χ_1	χ_2
$(0,0)$	$1-\chi_1(\mathfrak{a})$	$1-\chi_2(\mathfrak{a})$
$(1,0)$	0	$1-\chi_2(\mathfrak{a})$
$(0,1)$	$1-\chi_1(\mathfrak{a})$	0
$(1,1)$	0	0

Table 4: The substitution φ of a tile (θ,κ,ψ) (left), and the bits of $\varphi'(\mathfrak{a})$ (right). Child $(1,0)$ is south-east and child $(0,1)$ is north-west.

A.2 The base set

Distinct triples of $\mathbb{O}$ give distinct tiles: the four thin types have distinct north endpoints; a line has equal north and south endpoints while a corner does not, and the four corners have distinct north endpoints; the five black types have distinct bit tuples.

thin	base tiles	bit rule	variants each	letters
a	4 corners	$\chi_1 = \chi_2$	2	8
a	8 red crossings	$\chi_1 \neq \chi_2$	2	16
a	40 other lines	none	4	160
b	4 corners	$\chi_1 = \chi_2 = 0$	1	4
c	4 red crossings	$\chi_1 = 0, \chi_2 = 1$	1	4
c	20 other lines	$\chi_1 = 0$	2	40
d	4 red crossings	$\chi_1 = 1, \chi_2 = 0$	1	4
d	20 other lines	$\chi_2 = 0$	2	40
			total	276

Table 5: The 276 letters of $\mathbb{A}$. Each black class $\mathsf{B}, \mathsf{C}, \mathsf{D}, \mathsf{E}$ admits exactly two of the four red crossings, which gives 8 red crossings with thin type a and 4 each with thin types c and d.

position	thin	thick	red crossing exactly when
$(2,1)$	d	$\langle \partial_S\theta,\ \mathsf{r}_1 \rangle$	$\theta \in \{\mathsf{b}, \mathsf{d}\}$
$(2,3)$	d	$\langle \partial_N\theta,\ \mathsf{r}_3 \rangle$	$\theta \in \{\mathsf{a}, \mathsf{c}\}$
$(1,2)$	c	$\langle \mathsf{r}_0,\ \partial_W\theta \rangle$	$\theta \in \{\mathsf{b}, \mathsf{c}\}$
$(3,2)$	c	$\langle \mathsf{r}_2,\ \partial_E\theta \rangle$	$\theta \in \{\mathsf{a}, \mathsf{d}\}$

Table 6: Four cells of the depth-two block $\varphi^2(\theta, \kappa, \psi)$, for every base tile. Position $(2,1)$ is child $(0,1)$ of child $(1,0)$; $(2,3)$ is child $(0,1)$ of child $(1,1)$; $(1,2)$ is child $(1,0)$ of child $(0,1)$; $(3,2)$ is child $(1,0)$ of child $(1,1)$.

Lemma A.1 (Thin phase)**.** *For every tiling f of $\mathbb{O}$ there is a unique coset $c + 2\mathbb{Z}^2$ such that the thin type of $f(c + 2v + \varsigma)$ is $\mathsf{a}, \mathsf{c}, \mathsf{d}, \mathsf{b}$ for $\varsigma = (0,0), (1,0), (0,1), (1,1)$.*

Proof. The east endpoints of the thin types are distinct, and so are the west endpoints, so the thin type of the east neighbour is determined: $\mathsf{a} \leftrightarrow \mathsf{c}$, $\mathsf{b} \leftrightarrow \mathsf{d}$. Likewise the north neighbour: $\mathsf{a} \leftrightarrow \mathsf{d}$, $\mathsf{b} \leftrightarrow \mathsf{c}$. These involutions commute, so the thin type at $u + (x_1, x_2)$ is obtained from that at u by applying the first x_1 times and the second x_2 times. The sites of thin type a form one coset of $2\mathbb{Z}^2$. □

Lemma A.2 (The substitution is sound)**.** *(a) For every $t \in \mathbb{O}$, $\varphi(t)$ is a legal pattern of tiles of $\mathbb{O}$. (b) For every pattern w of $\mathbb{O}$, $\varphi(w)$ is legal if and only if w is legal. (c) φ is injective on triples.*

Proof. (a) Let $t = (\theta, \kappa, \psi)$. Child $(0,0)$ is $(\mathsf{a}, \kappa, \psi)$, which lies in $\mathbb{O}$ because every pair (κ, ψ) allowed for some thin type is allowed for a. Child $(1,1)$ is a corner with black A. Child $(1,0)$ is the line $\langle \partial_S\theta, \partial_E\kappa \rangle$ with thin type c and black $\mathrm{se}(\psi) \in \{\mathsf{B}, \mathsf{C}\}$; it is excluded only if $\mathrm{se}(\psi) = \mathsf{B}$ and $\partial_S\theta = \mathsf{r}_0$, or $\mathrm{se}(\psi) = \mathsf{C}$ and $\partial_S\theta = \mathsf{r}_2$. But $\partial_S\theta = \mathsf{r}_0$ only for $\theta = \mathsf{d}$, where $\psi \in \{\mathsf{D}, \mathsf{E}\}$ and $\mathrm{se}(\psi) = \mathsf{C}$; and $\partial_S\theta = \mathsf{r}_2$ only for $\theta = \mathsf{b}$, where $\psi = \mathsf{A}$ and $\mathrm{se}(\psi) = \mathsf{B}$. Child $(0,1)$ is the line $\langle \partial_N\kappa, \partial_W\theta \rangle$ with thin type d and black $\mathrm{nw}(\psi) \in \{\mathsf{D}, \mathsf{E}\}$; it is excluded only if $\mathrm{nw}(\psi) = \mathsf{D}$ and $\partial_W\theta = \mathsf{r}_3$, which needs $\theta = \mathsf{b}$, $\psi = \mathsf{A}$, where $\mathrm{nw}(\psi) = \mathsf{E}$; or $\mathrm{nw}(\psi) = \mathsf{E}$ and $\partial_W\theta = \mathsf{r}_1$, which needs $\theta = \mathsf{c}$, $\psi \in \{\mathsf{B}, \mathsf{C}\}$, where $\mathrm{nw}(\psi) = \mathsf{D}$. Inside the block the thin pattern $\mathsf{a}, \mathsf{c}, \mathsf{d}, \mathsf{b}$ matches; the thick endpoints across the four internal edges are $\partial_E\kappa$, $\partial_N\kappa$, $\partial_S\theta = \partial_S F(\theta)$ and $\partial_W\theta = \partial_W F(\theta)$ on both sides; and the black bits match by the identities of the black maps.

(b) Let $w(v) = (\theta, \kappa, \psi)$ and $w(v + \mathbf{e}_1) = (\theta', \kappa', \psi')$. Two fine edges cross the seam between their blocks. The lower one has thin endpoints $\mathsf{g}_3, \mathsf{g}_3$, thick endpoints $\partial_E\kappa$ and $\partial_W\kappa'$, and black bits $\partial_E\psi$ and $\partial_W\psi'$. The upper one has thin endpoints $\mathsf{g}_1, \mathsf{g}_1$, thick endpoints $\partial_E F(\theta) = \partial_E\theta$ and

$\partial_W\theta'$, and black bits $0,0$. So both match if and only if the coarse edge matches. Vertical seams are the same: the left fine edge records $\partial_N\kappa = \partial_S\kappa'$ and $\partial_N\psi = \partial_S\psi'$, the right one $\partial_N\theta = \partial_S\theta'$.

(c) Child $(0,0)$ shows κ and ψ, and child $(1,1)$ shows $F(\theta)$; F is a bijection. □

Lemma A.3 (Parents)**.** *Let w be a legal pattern of $\mathbb{O}$ on $[0,2)^2$ with thin types* a, c, d, b *at the sites $(0,0)$, $(1,0)$, $(0,1)$, $(1,1)$. Then $w = \varphi(t)$ for exactly one triple t. If w extends to a legal pattern on*

$$[0,2)^2 \cup \{(0,-1),(1,-1),(-1,0),(-1,1)\},$$

then $t \in \mathbb{O}$.

Proof. $w(1,1)$ has thin type b, so it is a corner with black A; let $\theta = F^{-1}$ of it, and let κ, ψ be the thick and black types of $w(0,0)$. The tile $w(1,0)$ has thin type c, so it is a line with black B or C; its west and north thick endpoints are $\partial_E\kappa$ and $\partial_S\theta$, and its west bit is $\partial_E\psi$, so it is $(\mathsf{c}, \langle \partial_S\theta, \partial_E\kappa\rangle, \mathrm{se}(\psi))$. In the same way $w(0,1) = (\mathsf{d}, \langle \partial_N\kappa, \partial_W\theta\rangle, \mathrm{nw}(\psi))$. So $w = \varphi(t)$ with $t = (\theta, \kappa, \psi)$, unique by Lemma A.2(c).

Child $(0,0)$ in $\mathbb{O}$ leaves the 52 pairs (κ, ψ) of thin type a. The children $(1,0)$ and $(0,1)$ lie in $\mathbb{O}$ unless $\theta = \mathsf{b}$ and $\psi \neq \mathsf{A}$, or $\theta = \mathsf{d}$ and $\psi \in \{\mathsf{A}, \mathsf{B}\}$, or $\theta = \mathsf{c}$ and $\psi \in \{\mathsf{A}, \mathsf{E}\}$; for $\theta = \mathsf{a}$ the new atoms are green. Comparing with Table 3, the triples obtained that are not in $\mathbb{O}$ are of two kinds: $\theta = \mathsf{c}$, $\psi = \mathsf{D}$, κ a line with $\alpha' \neq \mathsf{r}_3$; and $\theta = \mathsf{d}$, $\psi = \mathsf{C}$, κ a line with $\alpha \neq \mathsf{r}_2$.

Suppose $t = (\mathsf{c}, \langle \alpha, \alpha'\rangle, \mathsf{D})$ with $\alpha' \neq \mathsf{r}_3$ and w extends below. Then $w(1,0)$ is the line $\langle \mathsf{g}_0, \alpha'\rangle$. The tile at $(1,-1)$ has thin type b, so it is a corner; its north endpoint is g_0, so it is $\angle\mathsf{d}$, with west endpoint r_3. The tile at $(0,-1)$ has thin type d, so it is a line with black D or E; its north bit equals the south bit 1 of D, so its black is D, and its east endpoint is r_3. But thin d with black D forbids the horizontal atom r_3. The second kind is excluded in the same way by extending to the left: the tile at $(-1,1)$ is $\angle\mathsf{d}$ with south endpoint r_2, and the tile at $(-1,0)$ has thin type c, black C and vertical atom r_2, which is forbidden. □

Proposition A.4 (Base desubstitution)**.** $\mathbb{O}$ *tiles. For every tiling f of $\mathbb{O}$ there are a unique coset $c + 2\mathbb{Z}^2$ and, for each representative c, a unique tiling z of $\mathbb{O}$ with $f(c + 2v + \varsigma) = \varphi(z(v))(\varsigma)$.*

Proof. Take the coset of Lemma A.1; no other coset has thin type a at all its sites. Each block $f|_{c+2v+[0,2)^2}$ has the thin pattern of Lemma A.3 and the tiling supplies the cells below and to the left, so it equals $\varphi(z(v))$ for a unique $z(v) \in \mathbb{O}$, and z is legal by Lemma A.2(b). For existence, $\varphi^m(t)$ is a legal pattern on $[0,2^m)^2$ for every m, and König's lemma gives a tiling. □

A.3 The ruler decoration

Lemma A.5 (Decorated substitution is sound)**.** *(a) For every $\mathfrak{a} \in \mathbb{A}$, $\varphi'(\mathfrak{a})$ is a legal pattern of $\mathbb{A}$. (b) For every pattern w of $\mathbb{A}$, $\varphi'(w)$ is legal if and only if w is legal. (c) φ' is injective.*

Proof. (a) The bases are legal by Lemma A.2. Child $(0,0)$ has thin type a and both bits complemented, which preserves (R3) and (R4). Child $(1,1)$ is a corner of thin type b with bits $(0,0)$. Child $(1,0)$ has thin type c and $\chi_1 = 0$; its line $\langle \partial_S\theta, \partial_E\kappa\rangle$ is a red crossing only if $\partial_S\theta$ is red, that is $\theta \in \{\mathsf{b}, \mathsf{d}\}$, and then (R2) for $\mathfrak{a}$ gives $\chi_2(\mathfrak{a}) = 0$, so the child has bits $(0,1)$ and satisfies (R4). Child $(0,1)$ is symmetric. Inside the block the column bits are $1-\chi_1, 1-\chi_1$ and $0,0$ and the row bits $1-\chi_2, 1-\chi_2$ and $0,0$, so they match.

(b) Across a vertical seam between the blocks of $\mathfrak{a}$ and $\mathfrak{a}'$, the fine edges carry row bits $1-\chi_2(\mathfrak{a}), 1-\chi_2(\mathfrak{a}')$ and $0,0$; across a horizontal seam, column bits $1-\chi_1(\mathfrak{a}), 1-\chi_1(\mathfrak{a}')$ and $0,0$. With Lemma A.2(b) this gives the claim. (c) Child $(0,0)$ shows κ, ψ and the complemented bits, and child $(1,1)$ shows θ. □

Proof of Proposition 3.3. Soundness and injectivity are Lemma A.5. Let f be a tiling of $\mathbb{A}$, with base tiling $\bar{f}$; take c and $\bar{z}$ from Proposition A.4. The column bit of $f(x)$ depends only on x_1, say $X(x_1)$, and the row bit only on x_2, say $Y(x_2)$. A column $c_1 + 2v_1 + 1$ has thin types c, b, so $X = 0$ there by (R1); a row $c_2 + 2v_2 + 1$ has thin types d, b, so $Y = 0$ there. Put $z(v) = (\bar{z}(v), 1 - X(c_1 + 2v_1), 1 - Y(c_2 + 2v_2))$. Then $f(c + 2v + \varsigma) = \varphi'(z(v))(\varsigma)$, the bits of z are constant on columns and rows, and (R3), (R4) for $z(v)$ follow from those for $f(c + 2v)$. For (R1) and (R2) apply Proposition A.4 to $\bar{z}$: $\bar{z}(c' + 2m + \varsigma') = \varphi(\bar{w}(m))(\varsigma')$ for $\varsigma' \in [0,2)^2$, and with $o = c + 2c'$, $\bar{f}(o + 4m + y) = \varphi^2(\bar{w}(m))(y)$. By Table 6, the cells $(2,1)$ and $(2,3)$ of this block have thin type d, hence row bit 0, and exactly one of them is a red crossing, which then has column bit 1 by (R4); so $X(o_1 + 4m_1 + 2) = 1$. Likewise $Y(o_2 + 4m_2 + 2) = 1$. A letter $z(v)$ with $v = c' + 2m + \varsigma'$ has thin type b or c exactly when $\varsigma'_1 = 1$, and then $c_1 + 2v_1 = o_1 + 4m_1 + 2$ and $\chi_1(z(v)) = 0$: this is (R1). (R2) is the same. So z is a tiling of $\mathbb{A}$, and it is unique since its bases are and its bits are read off f. Taking $v \equiv c \pmod 2$ in $\{0,1\}^2$ and translating z gives the statement. Finally $\mathbb{A}$ tiles: $\varphi'^m(\mathfrak{a})$ is legal on $[0, 2^m)^2$ for every m. □

A.4 Ranks and red paths

Let f be a tiling of $\mathbb{A}$ with tower (f_m, v_m) and phase ϖ, and use the digits $\varsigma_m(x)$ of Section 3.3. The *rank* of a column x_1 is the least m with $\varsigma_{m,1}(x) = 1$, which by (1) is the 2-adic valuation of $x_1 - \varpi_1$; it is finite when $\varpi_1 \notin \mathbb{Z}$. Ranks of rows are defined with the second coordinate. For $r \geq 0$ put $s_i(z) = (z - \varpi_i) \bmod 2^{r+2}$.

Lemma A.6 (Ruler bits)**.** *If the column x_1 has rank r, then $\chi_1(f(x)) \equiv r \pmod 2$; rows are the same.*

Proof. A child ς has column bit 0 if $\varsigma_1 = 1$ and the complement of its parent's column bit if $\varsigma_1 = 0$ (Table 4). The letter $f_r(x^{(r)})$ has $\varsigma_{r,1}(x) = 1$, so its column bit is 0, and each of the r steps down to $f(x)$ complements it. □

A thick type with a red endpoint on the north–south axis has exactly one red atom value on that axis: for a line it is the vertical atom, at both its north and its south endpoint, and the corners $\angle\mathsf{b}$, $\angle\mathsf{c}$ have red north endpoints r_0, r_2 and $\angle\mathsf{a}$, $\angle\mathsf{d}$ red south endpoints r_0, r_2. On the east–west axis, $\angle\mathsf{b}$, $\angle\mathsf{a}$ have red east endpoints r_1, r_3 and $\angle\mathsf{c}$, $\angle\mathsf{d}$ red west endpoints r_1, r_3.

Lemma A.7 (Red paths)**.** *Let x_1 be a column of rank r. Then $s_1(x_1) \in \{2^r, 3 \cdot 2^r\}$, and for every $x = (x_1, x_2)$:*

(a) the thick type of $f(x)$ has a red endpoint on the north–south axis if and only if $s_2(x_2) \in [2^r, 3 \cdot 2^r]$;

(b) this endpoint is r_0 if $s_1(x_1) = 2^r$ and r_2 if $s_1(x_1) = 3 \cdot 2^r$;

(c) the thick type is a corner with red north endpoint if $s_2(x_2) = 2^r$, a corner with red south endpoint if $s_2(x_2) = 3 \cdot 2^r$, and a line if $2^r < s_2(x_2) < 3 \cdot 2^r$.

The same holds for a row x_2 of rank r with the coordinates exchanged: east–west red endpoints occur if and only if $s_1(x_1) \in [2^r, 3 \cdot 2^r]$, the endpoint is r_1 if $s_2(x_2) = 2^r$ and r_3 if $s_2(x_2) = 3 \cdot 2^r$, and the thick type is a corner with red east endpoint at $s_1(x_1) = 2^r$, a corner with red west endpoint at $s_1(x_1) = 3 \cdot 2^r$, and a line in between.

Proof. By (1), $s_1(x_1) = 2^r + 2^{r+1}\varsigma_{r+1,1}(x)$. We argue by induction on r, for all tilings at once. Let $r = 0$. The letter $f(x)$ is child $(1, \varsigma_{0,2})$ of $P = f_1(x^{(1)})$, which is child $\varsigma_1(x)$ of $f_2(x^{(2)})$; so P

has thin type $\theta = \mathsf{a},\mathsf{c},\mathsf{d},\mathsf{b}$ for $\varsigma_1(x) = (0,0),(1,0),(0,1),(1,1)$, and $f(x)$ has thick type $\langle \partial_{\mathrm{S}}\theta, \cdot\rangle$ if $\varsigma_{0,2} = 0$ and $F(\theta)$ if $\varsigma_{0,2} = 1$. With $s_2 = \varsigma_{0,2} + 2\varsigma_{1,2}$ and $s_1 = 1 + 2\varsigma_{1,1}$, Table 2 gives

$s_2(x_2)$	0	1	2	3
$s_1(x_1) = 1$	$\partial_{\mathrm{S}}\mathsf{a} = \mathsf{g}_2$	$F(\mathsf{a}) = \angle\mathsf{b}$	$\partial_{\mathrm{S}}\mathsf{d} = \mathsf{r}_0$	$F(\mathsf{d}) = \angle\mathsf{a}$
$s_1(x_1) = 3$	$\partial_{\mathrm{S}}\mathsf{c} = \mathsf{g}_0$	$F(\mathsf{c}) = \angle\mathsf{c}$	$\partial_{\mathrm{S}}\mathsf{b} = \mathsf{r}_2$	$F(\mathsf{b}) = \angle\mathsf{d}$

which is (a)–(c) for $r = 0$. Let $r \geq 1$, so $\varsigma_{0,1}(x) = 0$. The tiling f_1 has phase ϖ' with $\varpi = v_0 + 2\varpi'$; in it the column of $x^{(1)}$ has rank $r-1$, and with $s'_i(z) = (z - \varpi'_i) \bmod 2^{r+1}$ we have $s_i(x_i) = 2s'_i(x_i^{(1)}) + \varsigma_{0,i}(x)$. The letter $f(x)$ is child $(0, \varsigma_{0,2})$ of $P = f_1(x^{(1)})$, with thick type κ_P. If $\varsigma_{0,2} = 0$, $f(x)$ has thick type κ_P, $s_2 = 2s'_2$, and (a)–(c) follow from the induction hypothesis. If $\varsigma_{0,2} = 1$, s_2 is odd and $f(x)$ is the line $\langle \partial_{\mathrm{N}}\kappa_P, \cdot\rangle$; this endpoint is red exactly when κ_P is a line with a red vertical atom or a corner with red north endpoint, that is when $s'_2 \in [2^{r-1}, 3\cdot 2^{r-1})$, and the odd numbers $2s'_2 + 1$ with this property are exactly the odd numbers in $[2^r, 3\cdot 2^r]$, all strictly between 2^r and $3\cdot 2^r$. In both cases the atom is that of P, which gives (b). Rows are the same, with children $(\varsigma_{0,1}, 1)$ and $(1,0)$ and the endpoints $\partial_{\mathrm{W}}\theta$, $\partial_{\mathrm{E}}\kappa_P$. □

Proof of Lemma 3.14. Write $Y(\xi) = f^{\mathrm{q}}(\xi - \varepsilon)$ with phase ϖ of f, so $\tau = \varepsilon + 2\varpi$, and $\xi - \varepsilon = 2x + \varsigma$. Let r be the rank of the column x_1, finite since $\tau_1 \notin \mathbb{Z}$, and put $\hat{s}_i = (\xi_i - \tau_i) \bmod 2^{r+3}$, so that $\hat{s}_i = 2s_i(x_i) + \varsigma_i$. By Lemma A.7 and Definition 3.11, the quarter lies on a vertical red path if and only if $s_2(x_2) \in [2^r, 3\cdot 2^r]$, $\varsigma_1 = 1$ if $s_1(x_1) = 2^r$ and $\varsigma_1 = 0$ if $s_1(x_1) = 3\cdot 2^r$, and $\varsigma_2 = 1$ if $s_2(x_2) = 2^r$ and $\varsigma_2 = 0$ if $s_2(x_2) = 3\cdot 2^r$. That is,

$$\hat{s}_1 \in \{2^{r+1} + 1,\ 3\cdot 2^{r+1}\} \quad \text{and} \quad \hat{s}_2 \in [2^{r+1} + 1,\ 3\cdot 2^{r+1}],$$

with a left side exactly when $\hat{s}_1 = 2^{r+1} + 1$. By Lemma A.6 the path is light if and only if r is odd. For $r = 2d-1$, $2^{r+1} = 4^d$ and $2^{r+3} = 4^{d+1}$, and $\hat{s}_i \equiv \xi_i - a_d^{(i)} \pmod{4^{d+1}}$; the conditions say that ξ_1 is an endpoint of a column frame of depth d, the left one exactly for a left side, and that ξ_2 lies in a row frame of depth d. Conversely, if ξ_1 is an endpoint of a column frame of depth d, then $\xi_1 - \tau_1 \equiv 4^d + 1$ or $3\cdot 4^d \pmod{4^{d+1}}$, so $\varsigma_1 = 1$ and $x_1 - \varpi_1 \equiv 2^{2d-1}$, or $\varsigma_1 = 0$ and $x_1 - \varpi_1 \equiv 3\cdot 2^{2d-1}$, modulo 2^{2d+1}; in both cases the column has rank $2d-1$ and the first part applies. Light horizontal walls are the same, with the row half of Lemma A.7. □

A.5 Frames and laminar families

Proof of Lemma 3.13(a)–(c). (a) $I_d(m)$ lies in $[a_d + 4m\cdot 4^d, a_d + 4(m+1)4^d)$, and these windows are disjoint. Let $e < d$. Then $a_d \equiv a_e \pmod{4^{e+1}}$ and 4^{e+1} divides 4^d, so the endpoints of a frame of depth d are congruent to $a_e + 1$ and a_e modulo 4^{e+1}. An integer z lies in a frame of depth e exactly when $(z - a_e) \bmod 4^{e+1} \in [4^e + 1, 3\cdot 4^e]$, which excludes 0 and 1. So if a frame of depth e meets a frame of depth d, it contains neither endpoint of it and lies in its interior.

(b) Let $I_d(m) = [L, L + 2\cdot 4^d - 1]$. Left endpoints of frames of depth e form one residue class modulo 4^{e+1}, so one of them, q, lies in $[L, L + 4^{e+1} - 1]$, and its frame ends at $q + 2\cdot 4^e - 1 \leq L + 4^{e+1} + 2\cdot 4^e - 2 < L + 2\cdot 4^d - 1$.

(c) Call $s \in [0, 4^d)$ *allowed* if $s \bmod 4^{e+1} \notin [4^e + 1, 3\cdot 4^e]$ for every $1 \leq e < d$, and let g_d be the number of allowed residues; an integer z lies in no frame of depth below d exactly when $(z - a_d) \bmod 4^d$ is allowed. Then $g_1 = 4$. Writing $s \in [0, 4^{d+1})$ as $s = t\cdot 4^d + s'$ with $t \in [0,4)$, the conditions for $e < d$ concern s' and the condition for $e = d$ is $s \notin [4^d + 1, 3\cdot 4^d]$: for $t = 0$ all g_d allowed s' remain, for $t = 1$ only $s' = 0$, for $t = 2$ none, for $t = 3$ all except

$s' = 0$. Since 0 is allowed, $g_{d+1} = 2g_d$ and $g_d = 2^{d+1}$. Now $z = a_d + 4m \cdot 4^d + t \cdot 4^d + s'$ lies in $\mathring{I}_d(m)$ exactly when $t = 1$ and $s' \geq 2$, or $t = 2$. The residues 0 and 1 are allowed, so $|\mathcal{F}_d(m)| = (g_d - 2) + g_d = 2^{d+2} - 2$. □

Lemma A.8 (Laminar families)**.** *Let $\mathcal{J}$ be a set of integer intervals with at least two points each, such that any two distinct members are disjoint or one lies in the interior of the other, and let E be the set of their endpoints. Every $e \in E$ is an endpoint of exactly one member. If $w_1 < w_2$ are consecutive elements of E, w_1 a left endpoint and w_2 a right endpoint, then $[w_1, w_2] \in \mathcal{J}$. Conversely, the endpoints of a member that contains no other member are consecutive in E.*

Proof. If e were an endpoint of two members, they would meet, so one would lie in the interior of the other and would not have e as an endpoint. Let $w_1 = \min I$ and $w_2 = \max I'$ be consecutive, and suppose $I \neq I'$. If they are disjoint, then $\max I$ lies strictly between w_1 and w_2; if I lies in the interior of I', so does $\max I$; if I' lies in the interior of I, then $\min I'$ does. Each case contradicts consecutiveness. Conversely, let I contain no other member and $I' \neq I$: if I' meets I, then I lies in the interior of I', so the endpoints of I' lie outside I; otherwise no endpoint of I' lies in I. □

B Tables of the interaction

This appendix collects the fields, colours and components of Section 4 in tabular form. Together with the rules (O1)–(P3), (C0)–(C9) and the charge of Section 4, and the program of Section C, they define Φ. A site value is allowed when it satisfies every rule; neighbouring site values must have equal colours on their common edge.

field	values	in colours	rules
$\mathfrak{q}$	$\mathbb{A}^{\mathfrak{q}}$, 1104 quarters	all four, as $\mathfrak{q}_X$	walls (Definition 3.11)
$o^{\mathrm{W}}, o^{\mathrm{E}}$	$\{\mathrm{L}, \mathrm{R}\}^2$	c_{W}, c_{E}	(O1)
$o^{\mathrm{S}}, o^{\mathrm{N}}$	$\{\mathrm{B}, \mathrm{T}\}^2$	c_{S}, c_{N}	(O2)
$e^{\mathrm{W}}, e^{\mathrm{E}}, e^{\mathrm{S}}, e^{\mathrm{N}}$	$\{\mathrm{cell}, \mathrm{wall}\}^2 \cup \{*\}$	matching side	(E1)–(E3)
$\ell^{\mathrm{W}}, \ell^{\mathrm{E}}$	$\{\circ\} \cup \Lambda_{\mathrm{h}}^{+}$	c_{W}, c_{E}	(K1), (C0)–(C7)
$\ell^{\mathrm{S}}, \ell^{\mathrm{N}}$	$\{\circ\} \cup \Lambda_{\mathrm{v}}^{+}$	c_{S}, c_{N}	(K2), (C0)–(C8)
$\mathfrak{k}$	$\{\circ\} \cup \mathcal{K}$	none	(K3), (C0)–(C9)
$\mathfrak{r}$	$\{0, 1\}$	c_{W}, c_{E}	(Q1), (Q2), (P1)
$\mathfrak{c}$	$\{0, 1\}$	c_{S}, c_{N}	(Q1), (Q2), (P2)
σ	$\{+1, -1\}$	none	(P1)–(P3)
$w^{\mathrm{W}}, w^{\mathrm{E}}$	$\{+1, -1, \square\}$	c_{W}, c_{E}	(P1), (P3)
$w^{\mathrm{S}}, w^{\mathrm{N}}$	$\{+1, -1, \square\}$	c_{S}, c_{N}	(P2), (P3)

Table 7: The fields of a site value. The payload consists of σ, the wire values, the components x, y of $\mathfrak{k}$ and the components x of $\ell^{\mathrm{S}}, \ell^{\mathrm{N}}$; everything else is background.

Size. The site alphabet is the product of the value sets of Table 7, with $|\Lambda_{\mathrm{h}}^{+}| = 32|\mathcal{S}_M|$, $|\Lambda_{\mathrm{v}}^{+}| = 10368|\mathcal{S}_M|^2$ and $|\mathcal{K}| = 2^{17} \cdot 81\,|\mathcal{S}_M|(|\mathcal{S}_M| + 1)^2$, where $|\mathcal{S}_M| = |\Gamma_M|(|Q_M| + 1)$, $|\Gamma_M| = 96|\Gamma_U|$ and $|Q_M| = 54 + q_U$, with q_U the number of non-halting states of U. It is finite and depends only on U. We do not use its size.

component	colour	flows	meaning and rule
ct	horizontal	east	counter bit of the cell (C1)
tb	horizontal	east	next cell content $f_M(t_S, t_C, t_N)$ (C5)
hot	horizontal	west	one-hot marker, from the last column (C4)
rdy	horizontal	west	readiness, from the last column (C6)
g^-, g^+	horizontal	west	masks, from the last column (C7)
cy	vertical	north	counter carry (C1)
ay	vertical	north	adder carry, last column only (C3)
rc	vertical	north	row counter, saturating at 80 (C2)
rdy	vertical	north	readiness up the last column (C6)
tb	vertical	both	overlapping pair of cell contents (C5)
x	vertical	north	switches up the first column, payload (C8)

Table 8: Components of the logical colours Λ_h^+ and Λ_v^+. A component flowing east is written on the east colour and read on the west colour of the next cell; the others likewise.

component	values	role
$\mathrm{ct}_{\mathrm{in}}$, ct, cy	$\{0,1\}$	static binary counter: column i holds i (C1)
rc	$[0,80]$	logical row index, saturating at 80 (C2)
ay, sm	$\{0,1\}$	adds 3 to the last column, giving 2^{d+2} (C3)
hot	$\{0,1\}$	1 exactly on the logical row $d+2$ (C4)
t_S, t_N	$\mathcal{S}_M \cup \{\perp\}$	contents of the tape cells below and above, at the current time (C5)
t_C	$\mathcal{S}_M$	content of the tape cell at the current time (C5)
rdy	$\{0,1\}$	1 if M ended in q_{acc} at cell 0 (C6)
g^-, g^+	$\{0,1\}$	masks of lengths $20n-5$ and $20n+5$ on ready grids (C7)
x	$\{0,1\}^4$	switches of the two sensors, first column only (C8)
y	$\{0,1\}^4$	free bits of the two sensors, first column only (C9)

Table 9: Components of a cell record $\mathfrak{k} \in \mathcal{K}$, carried by active sites only.

C The machine on the grids

This appendix defines the machine M, its local rule f_M and the map init used in rules (C5)–(C7), and proves Lemma 5.5. The universal machine U of Section 2 has states Q_U, initial state q_U^0, halting states, tape alphabet Γ_U with blank $\sqcup$ and the symbol 1, and transition function δ_U; it never moves left from cell 0.

C.1 Tape, states and program

The tape of M has the cells $0, \dots, N-1$ of one logical column of a grid; cell j is the logical row j, and the time is the logical column. A tape symbol of M is a tuple

$$(\mathsf{h}, \mathsf{z}, \mathsf{u}, \mathsf{p}, \mathsf{n}, \mathsf{m}^-, \mathsf{m}^+) \in \Gamma_M = \{0,1\} \times \{0,1\} \times \Gamma_U \times \{0,1\} \times \{0,1,2\} \times \{0,1\} \times \{0,1\}.$$

The tracks h (the one-hot marker) and z (the mark of cell 0) are never written. The track u carries the tape of U, the track p marks used to compute $v_2(d)$, the track n the label in unary, and $\mathsf{m}^\pm$ the two masks. The set of cell contents is $\mathcal{S}_M = \Gamma_M \times (Q_M \cup \{\cdot\})$, where $\cdot$ means that the head is elsewhere, and

$$\mathrm{init}(h, z) = \big((h, z, \sqcup, 0, 0, 0, 0),\, q\big), \qquad q = q_{\mathrm{start}} \text{ if } z = 1, \quad q = \cdot \text{ if } z = 0.$$

The mask bits $\mathsf{m}^\pm(s)$ of $s \in \mathcal{S}_M$ are the tracks $\mathsf{m}^\pm$ of its tape symbol.

quantity	definition
VW, HW	side of a light vertical or horizontal wall at the quarter, or $-$ (Definition 3.11)
horizontal carrier	VW $= -$ and $o^{\mathrm{W}} = (\mathrm{L}, \mathrm{R})$
vertical carrier	HW $= -$ and $o^{\mathrm{S}} = (\mathrm{B}, \mathrm{T})$
role	active (both), horizontal, vertical, or inert (neither)
$f_{\mathrm{W}}, f_{\mathrm{E}}, f_{\mathrm{S}}, f_{\mathrm{N}}$	at active sites: $[e^{\mathrm{W}}_{-} = \text{wall}]$, $[e^{\mathrm{E}}_{+} = \text{wall}]$, $[e^{\mathrm{S}}_{-} = \text{wall}]$, $[e^{\mathrm{N}}_{+} = \text{wall}]$
root	active, $f_{\mathrm{W}} = 1$ and $f_{\mathrm{S}} = 1$
charge	at active sites with $f_{\mathrm{W}} = 1$ and $\mathrm{rc} \le 79$: the switch $x_{\lfloor \mathrm{rc}/20 \rfloor}$; otherwise 0

Table 10: Quantities read from one site value.

The states of M are $P_0, \dots, P_{10}$ with $q_{\text{start}} = P_0$; $S_1, \dots, S_6$; H_0, H_1; the pairs (q, run) for the non-halting states q of U; E_0, E_1, $E_2^{(c)}$ for $c \in [1, 20]$, E_3, $E_4^{(c)}$ for $c \in [1, 4]$, $E_5^{(c)}$ for $c \in [1, 5]$, E_6; and q_{acc}, q_{fail}. At each step M reads the symbol under its head, may change some tracks, moves up ($+1$), down (-1) or stays (0), and changes state, according to Table 11. Every pair of a state and a symbol not covered by the table, and every attempt to move down from a cell with $\mathsf{z} = 1$, leads to q_{fail} without writing or moving. The states q_{acc} and q_{fail} are absorbing: the head stays and nothing is written.

C.2 The local rule

For a cell content $s = (g, q)$ with a head ($q \neq \cdot$) let $\delta_M(q, g) = (q', g', D)$ be the step of Table 11, with the convention for uncovered pairs. The local rule $f_M(a, b, c)$, for $a, c \in \mathcal{S}_M \cup \{\perp\}$ and $b = (g_b, q_b) \in \mathcal{S}_M$, is defined by the first applicable case:

(1) if $q_b \neq \cdot$ and $\delta_M(q_b, g_b) = (q', g', D)$: return (g_b, q_{fail}) if $D = +1$ and $c = \perp$, or if $D = -1$ and $a = \perp$; otherwise return (g', q') if $D = 0$ and $(g', \cdot)$ if $D \neq 0$;

(2) if $a = (g_a, q_a)$ with $q_a \neq \cdot$ and $\delta_M(q_a, g_a) = (q', g', +1)$: return (g_b, q');

(3) if $c = (g_c, q_c)$ with $q_c \neq \cdot$ and $\delta_M(q_c, g_c) = (q', g', -1)$: return (g_b, q');

(4) otherwise return $(g_b, \cdot)$.

This is a total function on a finite set; it is the usual radius-one cellular-automaton form of a Turing machine.

Definition C.1 (The run)**.** For $d \ge 1$ and $N = N_d$, the *run* of M on the grid of depth d is the sequence of maps $c_i \colon [0, N) \to \mathcal{S}_M$, $i \in [0, N)$, with $c_0(j) = \mathrm{init}([j = d + 2], [j = 0])$ and $c_{i+1}(j) = f_M(c_i(j-1), c_i(j), c_i(j+1))$, where $c_i(-1) = c_i(N) = \perp$.

Since $d + 2 < N_d$, the column c_0 has $\mathsf{h} = 1$ exactly at cell $d + 2$, $\mathsf{z} = 1$ exactly at cell 0, and one head, at cell 0 in state P_0. By induction on i, each c_i has exactly one cell with a head, and c_{i+1} is obtained from c_i by one step of M, except that a step that would move the head off the tape $[0, N)$ instead puts it in q_{fail}. So (c_i) is the run of M for $N - 1$ steps on a tape of N cells, with an exit through either end counted as failure.

C.3 Proof of the readiness lemma

Let $n = n(d) = 1 + v_2(d)$. We follow the program from c_0.

Depth. In state P_k the head is at cell k while $k < 10$, and at a cell at least 10 in state P_{10}. It moves up, marking $\mathsf{p} = 1$ at the cells $2, \dots, d + 1$, until it reads $\mathsf{h} = 1$ at cell $d + 2$. If $d + 2 \le 9$,

state	reads	writes	move	next state
P_k	$\mathsf{h} = 0$	$\mathsf{p} := 1$ if $k \geq 2$	$+1$	$P_{\min(k+1,10)}$
P_k, $k \leq 9$	$\mathsf{h} = 1$		0	q_{fail}
P_{10}	$\mathsf{h} = 1$		0	S_1
S_1	$\mathsf{z} = 0$ / $\mathsf{z} = 1$	– / $\mathsf{n} := 1$	-1 / 0	S_1 / H_0
H_π	$\mathsf{h} = 0, \mathsf{p} = 0$		$+1$	H_π
H_0	$\mathsf{h} = 0, \mathsf{p} = 1$		$+1$	H_1
H_1	$\mathsf{h} = 0, \mathsf{p} = 1$	$\mathsf{p} := 0$	$+1$	H_0
H_0 / H_1	$\mathsf{h} = 1$		0	S_2 / S_4
S_2	$\mathsf{z} = 0$ / $\mathsf{z} = 1$		-1 / 0	S_2 / S_3
S_3	$\mathsf{n} = 1$ / $\mathsf{n} = 0$	– / $\mathsf{n} := 1$	$+1$ / 0	S_3 / S_1
S_4	$\mathsf{z} = 0$ / $\mathsf{z} = 1$		-1 / 0	S_4 / S_5
S_5	$\mathsf{n} = 1$ / $\mathsf{n} = 0$	$\mathsf{u} := 1$ / –	$+1$ / 0	S_5 / S_6
S_6	$\mathsf{z} = 0$ / $\mathsf{z} = 1$		-1 / 0	S_6 / (q_U^0, run)
(q, run)	$\mathsf{u} = \gamma$, $\delta_U(q, \gamma) = (q', \gamma', D)$	$\mathsf{u} := \gamma'$	D	(q', run), or E_0 if q' halts
E_0	$\mathsf{z} = 0$ / $\mathsf{z} = 1$		-1 / 0	E_0 / E_1
E_1	$\mathsf{n} = 2$ / $\mathsf{n} = 1$ / $\mathsf{n} = 0$	– / $\mathsf{n} := 2$ / –	$+1$ / 0 / 0	E_1 / $E_2^{(20)}$ / E_3
$E_2^{(c)}$	$\mathsf{m}^- = 1$		$+1$	$E_2^{(c)}$
$E_2^{(c)}$, $c \geq 2$	$\mathsf{m}^- = 0$	$\mathsf{m}^- := 1$, $\mathsf{m}^+ := 1$	$+1$	$E_2^{(c-1)}$
$E_2^{(1)}$	$\mathsf{m}^- = 0$	$\mathsf{m}^- := 1$, $\mathsf{m}^+ := 1$	0	E_0
E_3	$\mathsf{m}^+ = 1$ / $\mathsf{m}^+ = 0$	– / $\mathsf{m}^+ := 1$	$+1$ / $+1$	E_3 / $E_4^{(4)}$
$E_4^{(c)}$, $c \geq 2$ / $E_4^{(1)}$	any	$\mathsf{m}^+ := 1$	$+1$ / -1	$E_4^{(c-1)}$ / $E_5^{(5)}$
$E_5^{(5)}$	$\mathsf{m}^- = 0$ / $\mathsf{m}^- = 1$	– / $\mathsf{m}^- := 0$	-1 / -1	$E_5^{(5)}$ / $E_5^{(4)}$
$E_5^{(c)}$, $2 \leq c \leq 4$	$\mathsf{m}^- = 1$	$\mathsf{m}^- := 0$	-1	$E_5^{(c-1)}$
$E_5^{(1)}$	$\mathsf{m}^- = 1$	$\mathsf{m}^- := 0$	0	E_6
E_6	$\mathsf{z} = 0$ / $\mathsf{z} = 1$		-1 / 0	E_6 / q_{acc}

Table 11: The program of M. Entries separated by "/" are alternative cases, matched in order. In the row for (q, run), $D = +1$ if U moves right and $D = -1$ if it moves left.

that is $d \leq 7$, it fails. Otherwise it enters S_1, which returns to cell 0, writes $\mathsf{n} = 1$ there and enters H_0.

Label. A pass in states H_0, H_1 scans the cells from 0 up to the marker, keeps the first, third, ... marked cell and erases the second, fourth, ...; it ends in H_0 if the number c of marks was even and in H_1 if it was odd. In the even case $c/2$ marks remain, and S_2, S_3, S_1 append one cell with $\mathsf{n} = 1$ above the existing ones and start the next pass at cell 0. Starting from $c = d \geq 8$, the passes end after $v_2(d)$ even passes and one odd pass, leaving $\mathsf{n} = 1$ exactly on the cells $0, \ldots, n-1$.

Simulation. S_4, S_5, S_6 write the symbol 1 on the u-track of the cells $0, \ldots, n-1$ and return to cell 0 in the state (q_U^0, run). From then on each step of M is one step of U on the u-track, started on 1^n, until U enters a halting state, at which point M enters E_0. If U does not halt on n, M never leaves the states (q, run) and q_{fail}.

Masks. E_0 returns to cell 0. Each round of $E_1, E_2^{(20)}, \ldots, E_2^{(1)}, E_0$ consumes the lowest cell with $\mathsf{n} = 1$, setting it to 2, and extends both masks by 20 cells; after n rounds E_1 reaches the cell n, where $\mathsf{n} = 0$, and $\mathsf{m}^- = \mathsf{m}^+ = 1$ exactly on the cells $0, \ldots, 20n-1$. Then E_3 and E_4 set $\mathsf{m}^+ = 1$ on the cells $20n, \ldots, 20n+4$, E_5 clears m^- on the cells $20n-5, \ldots, 20n-1$, and

E_6 returns to cell 0 and enters the absorbing state q_{acc}. Then $\mathsf{m}^- = 1$ exactly on the cells $j < 20n - 5$ and $\mathsf{m}^+ = 1$ exactly on the cells $j < 20n + 5$.

Proof of Lemma 5.5(a). The state q_{acc} is entered only from E_6, which is reached only through the whole program above; q_{fail} is absorbing. So if $c_{N-1}(0)$ carries q_{acc}, then $d \geq 8$, U halted on n, the masks are as stated and stay so, since q_{acc} writes nothing, and the head wrote at cell $20n + 4$ without leaving the tape, so $20n + 5 \leq N_d$.

Proof of Lemma 5.5(b). Suppose U halts on n, after $t_U(n)$ steps, with largest visited cell $s_U(n)$. Let $d \geq 8$ with $n(d) = n$, so $n \leq d$. Counting the steps of each phase: the depth phase and the return take at most $2(d + 3)$ steps; there are at most $v_2(d) + 1 \leq d$ passes, each followed by at most $2(d + 3) + 2(n + 2)$ steps; loading the input takes at most $2(d + 3) + 2(n + 1)$ steps; the simulation takes $t_U(n)$ steps; the return after it at most $s_U(n) + 1$; the n mask rounds at most $n(41n + 3)$; and the end at most $40n + 25$. With $n \leq d$ the total is at most $45d^2 + 59d + 40 + t_U(n) + s_U(n) \leq 50(d + 2)^2 + t_U(n) + s_U(n)$. The head never goes above the cell $\max(d + 2, s_U(n), 20n + 4)$. Since $N_d - 1 = 2^{d+2} - 3$, there is d_0 such that for all $d \geq d_0$ both bounds are below $N_d - 1$, and then the run reaches q_{acc} at cell 0 before the last column and stays there, so $\rho_d = 1$. The depths $d = 2^{n-1}(2k + 1)$, $k \geq 0$, all have label n, so infinitely many of them are at least d_0. □

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